

Engineering a Magnetometer using Diamond NV Centers

Yongwoo Park
Advised by Prof. Viva Horowitz, Ph. D

Submitted in accordance with the requirements for the degree of
Bachelor of Arts

Department of Physics
Hamilton College
December 6, 2023

Table of Contents

ABSTRACT.....	3
ACKNOWLEDGEMENTS	4
1 – THEORY	5
1.1 – NITROGEN VACANCY CENTERS	5
1.2 – GROUNDED COPLANAR WAVEGUIDES	8
1.3 – LOCK-IN AMPLIFICATION/CHOPPING	10
2 – PREVIOUS WORK.....	13
3 – APPARATUS	16
3.1 – PHOTOLUMINESCENCE DETECTION SYSTEM	16
3.2 – ANTENNA	20
3.3 – FOCUSING MICROSCOPE ONTO ANTENNA	23
3.4 – LOCK-IN CHOPPING.....	25
4 – ODMR PROGRESS	29
4.1 – USING SAME MEASUREMENT METHOD AS WRIGHT	29
4.2 – USING LOCK-IN CHOPPING.....	31
4.3 – ADDING AMPLIFIER TO SG380 SIGNAL.....	34
5 – RESULTS/CONCLUSION	37
6 – FUTURE WORK.....	38
REFERENCES.....	41
APPENDIX.....	42
A.....	42
B	45

Abstract

Diamond Nitrogen Vacancy (NV) Centers can be used as an extremely sensitive magnetic field sensor. The Diamond NV Center's quantum system is easily manipulable, and its current state can be detected by its fluorescence when excited by a laser. The purpose of this project was to create a system that can measure magnetic fields to a high resolution using the aforementioned properties of diamond NV centers. Using a confocal microscope and a microwave antenna, the fluorescence of the diamond NV center was recorded over a spectrum of microwave frequencies, and the resulting data was analyzed to find the magnetic field. The system can successfully measure magnetic fields; however, there are improvements and changes to be made to both the hardware and software to get more consistent results out of the system.

Acknowledgements

I would like to thank all of the physics professors at Hamilton. I would specifically like to thank Viva Horowitz, Gordon Jones, and my advisor Kate Brown for their support and the opportunities they provided me with, as well as the skills and material they taught me.

I would also like to thank my friends and classmates for collaborating with me on countless homework problems and for just being fun people to hang with. A special thanks to Lauren Kuster '23 and Kai Haesslein '24 for their work in the Horowitz lab and the knowledge they passed upon to me while we worked together over summer and after.

I additionally would like to thank my family for their support and motivating me to keep pushing. I would not have had the opportunity to get this education without them and their support throughout my life.

1 – Theory

1.1 – Nitrogen Vacancy Centers

Nitrogen Vacancy Centers (NV Centers) are a type of imperfection in the crystal lattice of a diamond. An NV center is characterized by a nitrogen atom replacing a carbon atom, known as a substitutional nitrogen atom, and an adjacent lattice vacancy.

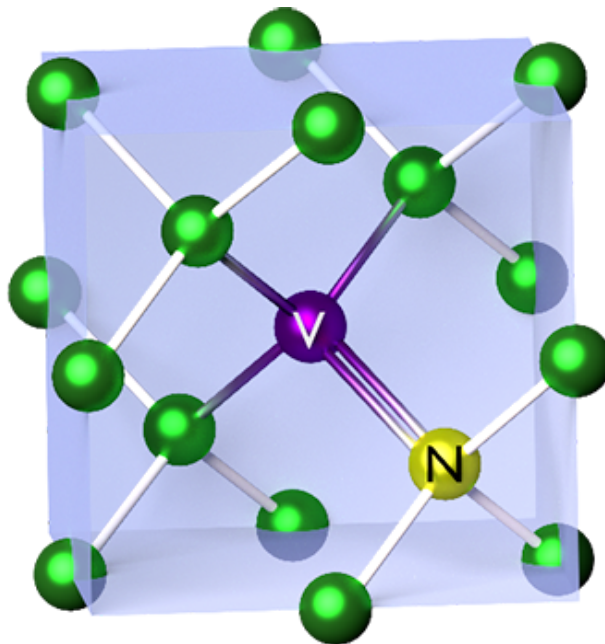


Figure 1.1 – Nitrogen Vacancy Center in a diamond crystal lattice. The green atoms are carbon, the yellow is the substitutional nitrogen atom, and the purple V is the vacancy. Reprinted from reference [1].

NV centers behave like molecules with a spin-1 quantum system. The NV Center's spin states are closely related to its fluorescence, and thusly they are useful for a wide range of applications. This section will explain how the NV centers' quantum system will be manipulated to be used as a magnetometer.

NV centers involve six electrons; 2 from the nitrogen atom, 3 from the surrounding carbon atoms, and 1 captured from the diamond lattice. The electronic levels are given by a ground state, an excited state, and a metastable singlet state. The ground state and excited states are spin triplets ($S=1$) that split into three spin sublevels ($m_s=0,\pm 1$). The NV centers fluoresce

when the electrons transition from the excited to ground state. Most wavelengths below the zero phonon line, 638 nm, can consistently excite the electrons from the ground to excited state. The photons released during relaxation are emitted at 630-800nm, with the difference in absorbed and released photon energy releasing as a phonon [2].

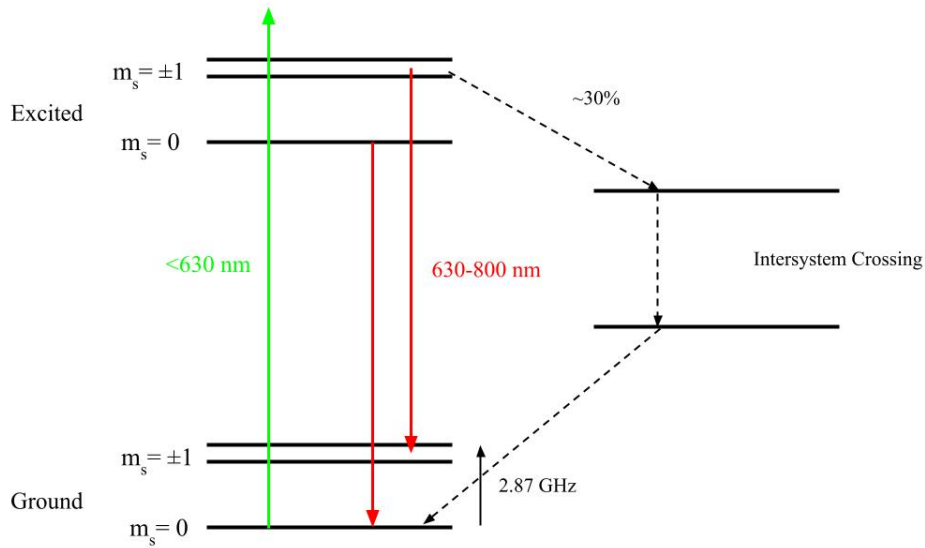


Figure 1.2 – Energy level diagram for a Diamond NV Center. The green arrow is the laser exciting the electrons, and the red arrows are the electrons relaxing and emitting photons. The black arrow is the transition caused by 2.87 GHz microwave radiation. The dotted arrows represent the transition through the ISC.

As a result of the NV center's axial symmetry, the $m_s = \pm 1$ states are degenerate and energetically higher than the $m_s = 0$ state. Microwave radiation at 2.87 GHz can excite an electron in the ground state from $m_s = 0$ to $m_s = \pm 1$. The $m_s = \pm 1$ states' energies shift in opposite directions in the presence of a magnetic field due to the Zeeman Effect – this is what allows us to use the NV centers for magnetic field sensing. The energy separation between the $m_s = -1$ and $m_s = +1$ states are given by $E = 2\gamma B_z$. The gyromagnetic ratio γ is given by $\gamma = 2\pi * 28 \text{ GHz/T}$, and B_z is the magnetic field component in the direction of the NV axis [2].

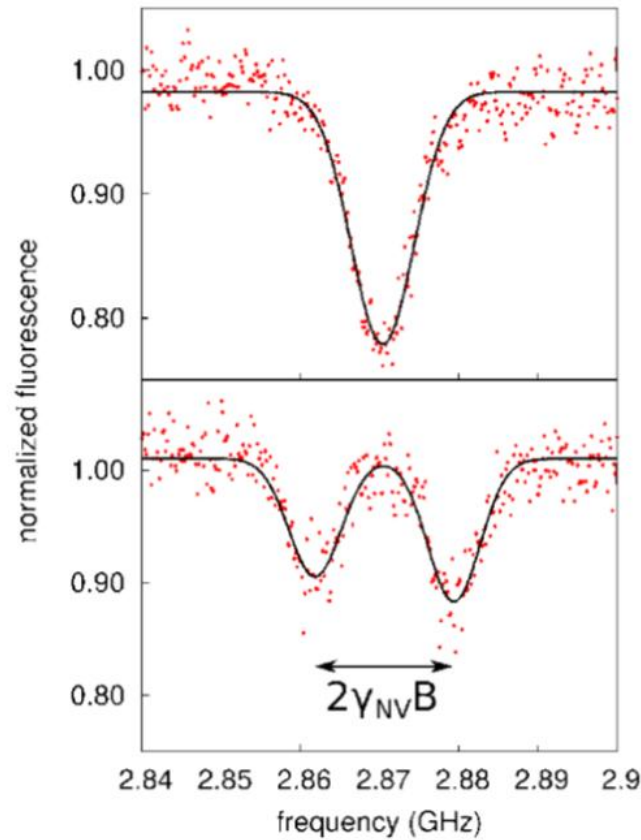


Figure 1.3 – ODMR Spectra for Diamond NV Centers. The upper spectrum is for an NV center without a magnetic field present, and the lower spectrum is for an NV center in a magnetic field. Reprinted from reference [3].

The NV center's optical electronic transitions are highly spin preserving, so the electrons' spin state does not change when transitioning between the ground and excited states. However, when transitioning from the excited $m_s=\pm 1$ state to the ground $m_s=\pm 1$ state, there is a 30% chance that the electron relaxes through the metastable singlet state (intersystem crossing), to the ground $m_s=0$ state [2]. This is useful for two reasons; first, this means that prolonged laser exposure sends all the electrons to the ground $m_s=0$ state. Second, electrons relaxing through the intersystem crossing do not release photons of visible wavelengths, and thus there is approximately a 30% difference in luminescence between the $m_s=0$ and $m_s=\pm 1$ transitions [2].

With these means of manipulating the NV Center's quantum system and luminescent indicators of the spin state, we can measure magnetic fields with NV centers. Reading out the

spin state of a quantum system using luminescence is known as Optically Detected Magnetic Resonance (ODMR).

1.2 – Grounded Coplanar Waveguides

As mentioned in the previous section, microwave radiation at 2.87 GHz excites electrons in the NV Center from the $m_s=0$ state to the $m_s=\pm 1$ state at zero magnetic field. To create this radiation, we will use a Grounded Coplanar Waveguide (GCPW). Another suitable option is a microstrip, but GCPW's are favorable for our applications on micro-and nano-diamonds because more signal is concentrated in the traces, where we can easily place the diamonds, while more of the microstrip's signal goes through the substrate.

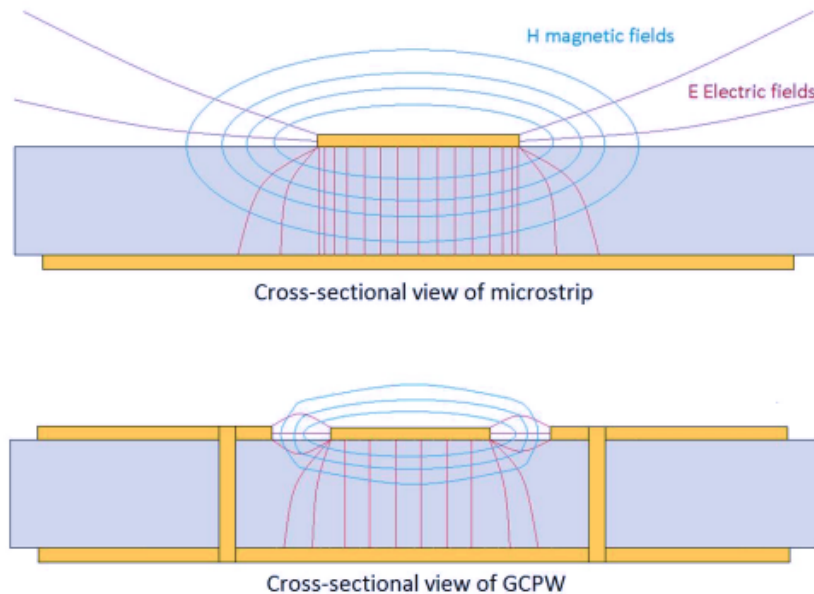


Figure 1.4 – Cross sectional diagram of microstrip and GCPW, and their respective magnetic and electric fields. Reprinted from reference [4].

The design and dimensions of our GCPW will be dependent on our desired characteristic impedance ($50\ \Omega$) to ensure there is minimal signal reflection and efficient signal transfer.

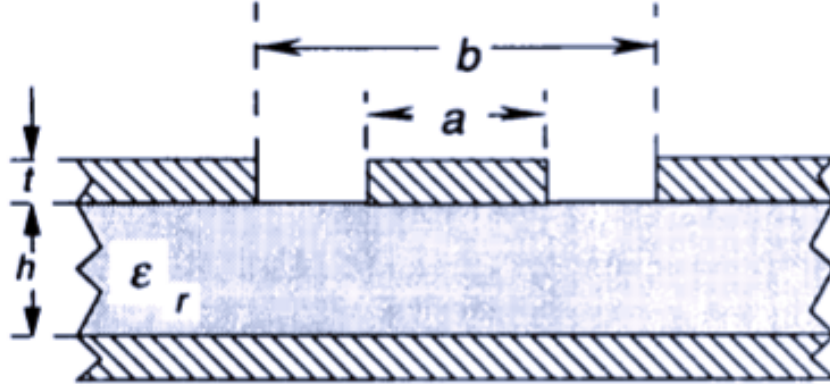


Figure 1.5 – Cross sectional diagram of GCPW, with labeled dimensions a , b , t , h . Reprinted from reference [5].

By using EM field theory to analyze the GCPW [5], the dimensions a , b , t , and h in figure 1.5 govern the characteristic impedance (Z_0) of the grounded coplanar waveguide. The characteristic impedance is given by

$$Z_0 = \frac{60\pi}{\sqrt{\epsilon_{eff}}} \frac{1}{\frac{K(k_1)}{K(k'_1)} + \frac{K(k_2)}{K(k'_2)}}$$

where ϵ_{eff} is the effective relative permittivity of the GCPW, given by

$$\epsilon_{eff} = \frac{1 + \epsilon_r \frac{K(k'_1)}{K(k_1)} \frac{K(k_2)}{K(k'_2)}}{1 + \frac{K(k'_1)}{K(k_1)} \frac{K(k_2)}{K(k'_2)}}$$

ϵ_r is the relative permittivity of the circuit board substrate, and k_1 , k_2 , k'_n are constants determined by the dimensions of the GCPW, given by

$$k_1 = \frac{a}{b}$$

$$k_2 = \frac{\tanh\left(\frac{\pi a}{4h}\right)}{\tanh\left(\frac{\pi b}{4h}\right)}$$

$$k'_n = \sqrt{1 - k_n^2} \quad n = 1, 2$$

and $K(k)$ is an elliptic integral [5]. Thus, we are able to solve for the GCPW's impedance computationally.

1.3 – Lock-in Amplification/Chopping

Lock-in Amplification is a technique that allows a weak and repetitive signal to be extracted from a background of noise. It works by taking a clean reference signal, usually a sine wave, with the same frequency as the signal to be extracted. Then, the noisy input signal and the reference signal are mixed by multiplying them; this creates a new signal known as the Homodyne signal.

Since the input signal can be expanded into a Fourier Series, the Homodyne signal is the result of multiplying two sinusoidal functions. When multiplied with the reference signal, components of the input signal with a different frequency than the reference frequency become functions that have a time average of zero due to the orthogonality of sinusoidal functions [8]. On the other hand, multiplying components of the input signal of the same frequency as the reference signal produces a sinusoidal wave at twice the frequency; furthermore, as long as the two signals are not 90° or 270° out of phase, the resulting function will not have a time average of zero. Depending on the phase offset, the resulting time average value will be different, and its magnitude will not exceed half the product of the reference and input amplitudes. Finding this nonzero time average is the mechanism by which the lock-in amplifier detects the desired component of the input signal [9].

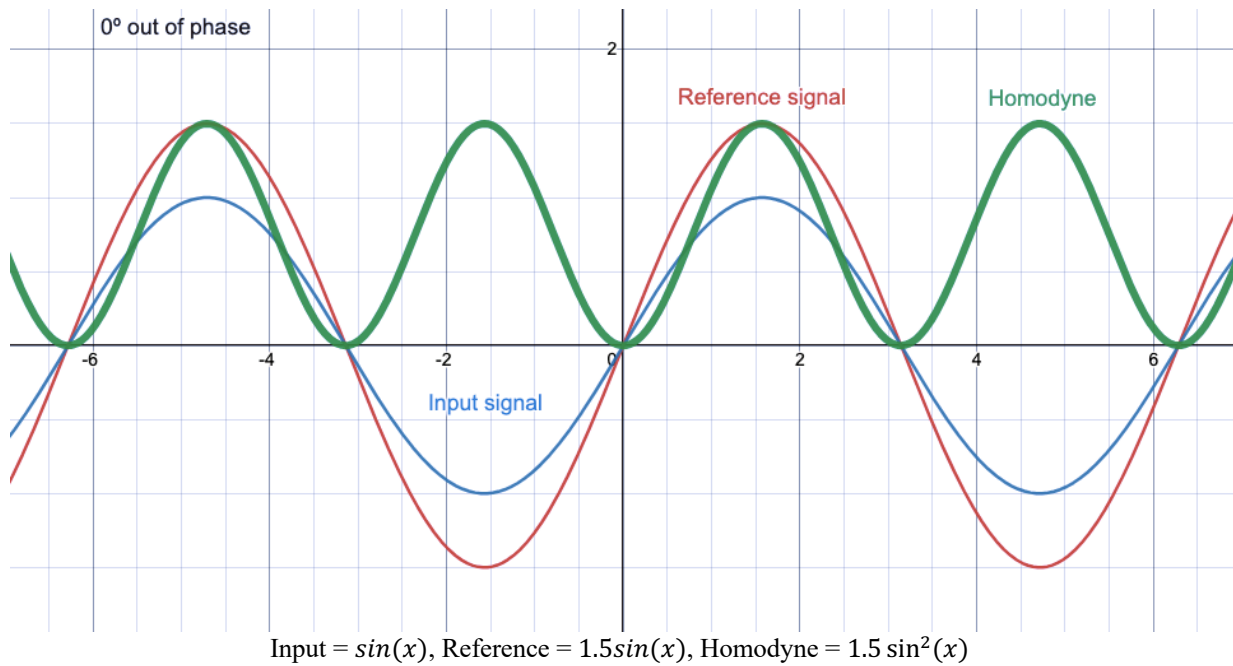


Fig. 1.5a – Here, the reference and input signals 0° are out of phase, with the same frequencies. This produces a Homodyne with a nonzero time average.

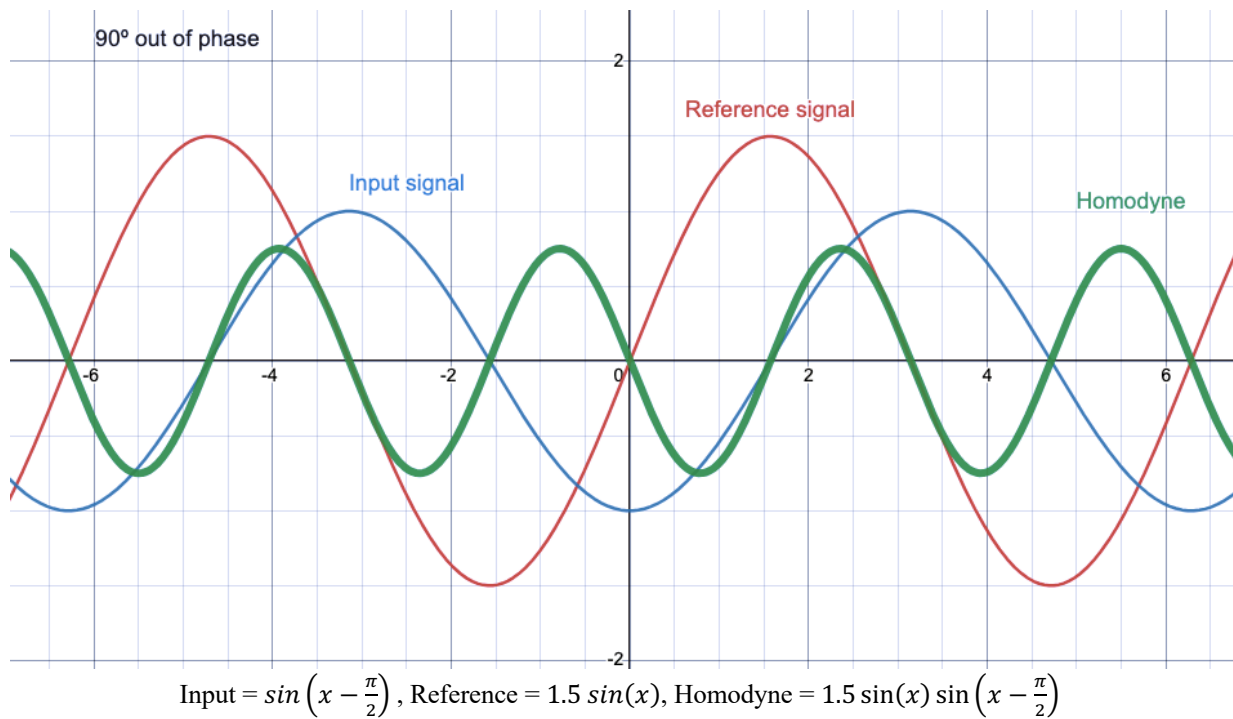


Fig. 1.5b – The reference and input signals are 90° out of phase, with the same frequencies, producing a Homodyne with a time average of zero.

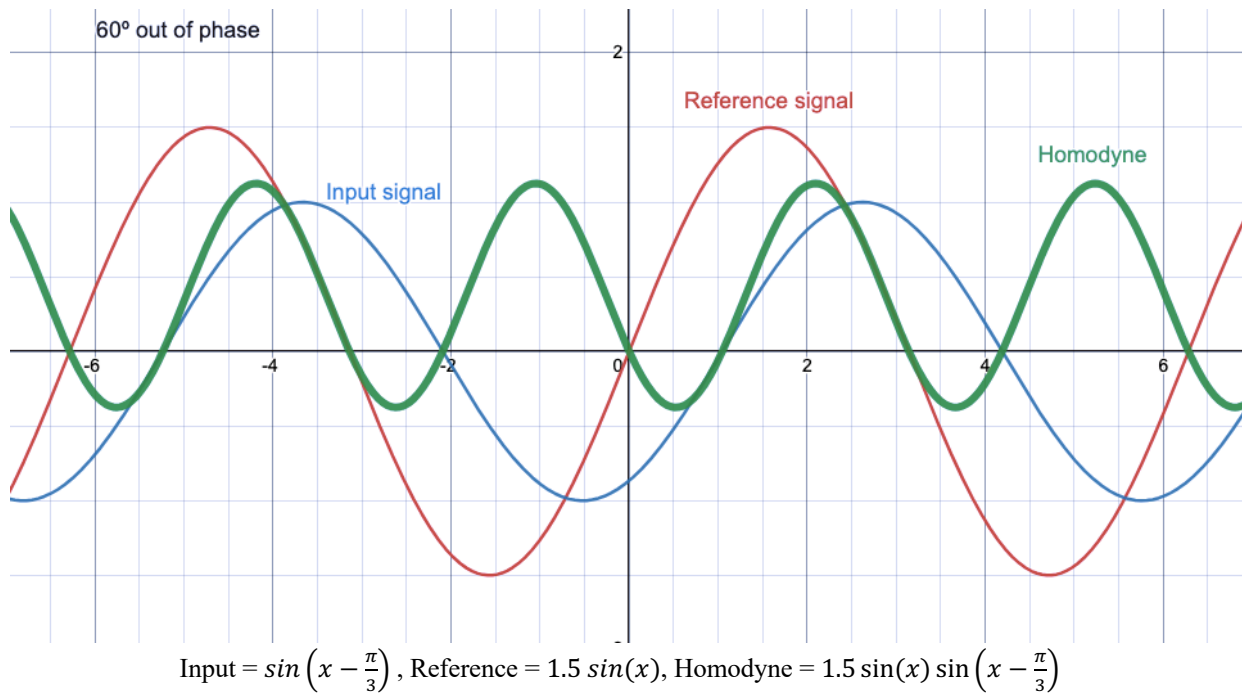


Fig. 1.5c – The reference and input signals are 60° out of phase, with the same frequencies, producing a Homodyne with nonzero time average. This Homodyne's time average is less than 1.5a's time average.

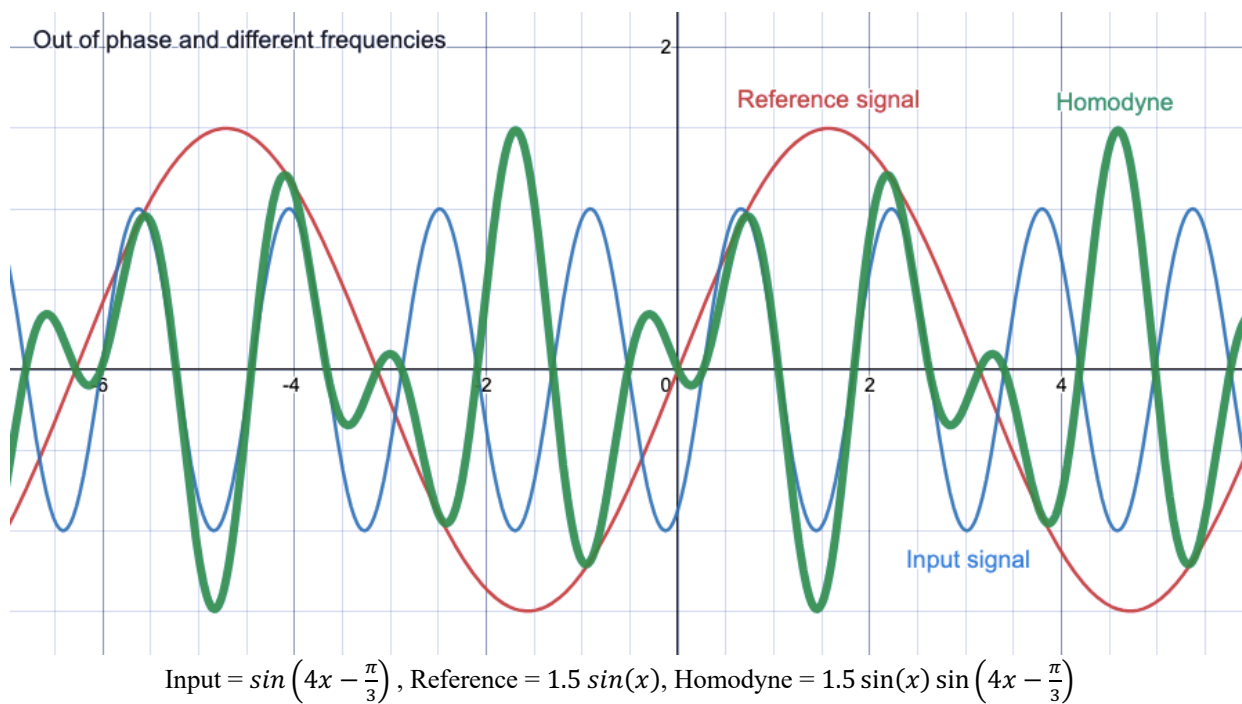


Fig 1.5d – Reference and input signals are out of phase and different frequencies, producing a Homodyne with a time average of zero. Any reference and input signals with different frequencies will produce a Homodyne with a time average of zero.

Figure 1.5a, 1.5b, 1.5c, 1.5d – Graphical representations of Homodyne signals with the input signal and reference signals that produced them. The input and reference signals are sine waves, and the Homodyne is the product of the input and reference signals.

To remove noise and unwanted trends from our data, a simpler alternative to lock-in amplification was employed, which will be referred to as “Lock-in Chopping”; this alternative used a square wave as the reference signal, and a switch. The switch takes an input signal, a reference signal, and has two outputs. The input signal is sent through to one of its two outputs, and which output it is sent through is determined by whether the square wave reference signal is high or low.

At the same time, the same square wave modulates the signal being sent to the GCPW; there is only signal going to the GCPW when the square wave is high. Thus, the measurements taken from one output of the switch are all control measurements, while the measurements taken through the other output are in the presence of microwave radiation from the GCPW. The difference between these measurements gives us an accurate picture of the change in luminosity caused by microwave radiation among any externally caused patterns of change in the luminosity.

The signal extracted by Lock-in Chopping does not require syncing a reference signal and input signal’s phase, because the microwave turns on and off at the same time that the measurements switch from A to B, while Lock-in Amplification still returns a signal for out-of-phase input signals (see fig. 1.5c), which we are not looking for.

2 – Previous Work

This section will outline the previous work done by Lucas Wright ’21 on ODMR in the Horowitz lab, and work done by Kai Haesslein ’24 in Oregon during summer 2023.

Using an SG380 Series RF Signal Generator controlled by a python program (see appendix A), a 2.4/5.8GHz 10 dBi Dual Polarity MIMO Panel Antenna, a 532 nm laser, an Avalanche Photodiode (APD), and an SR400 photon counter, Wright produced a frequency vs luminescence spectrum that gave evidence of a changing electron spin state in the NV center from $m_s = 0$ to $m_s = \pm 1$. Wright stated that more measurements of the spectrum needed to be made to find the shape of the ODMR spectrum but was unable to replicate the measurement for “an unknown reason.”

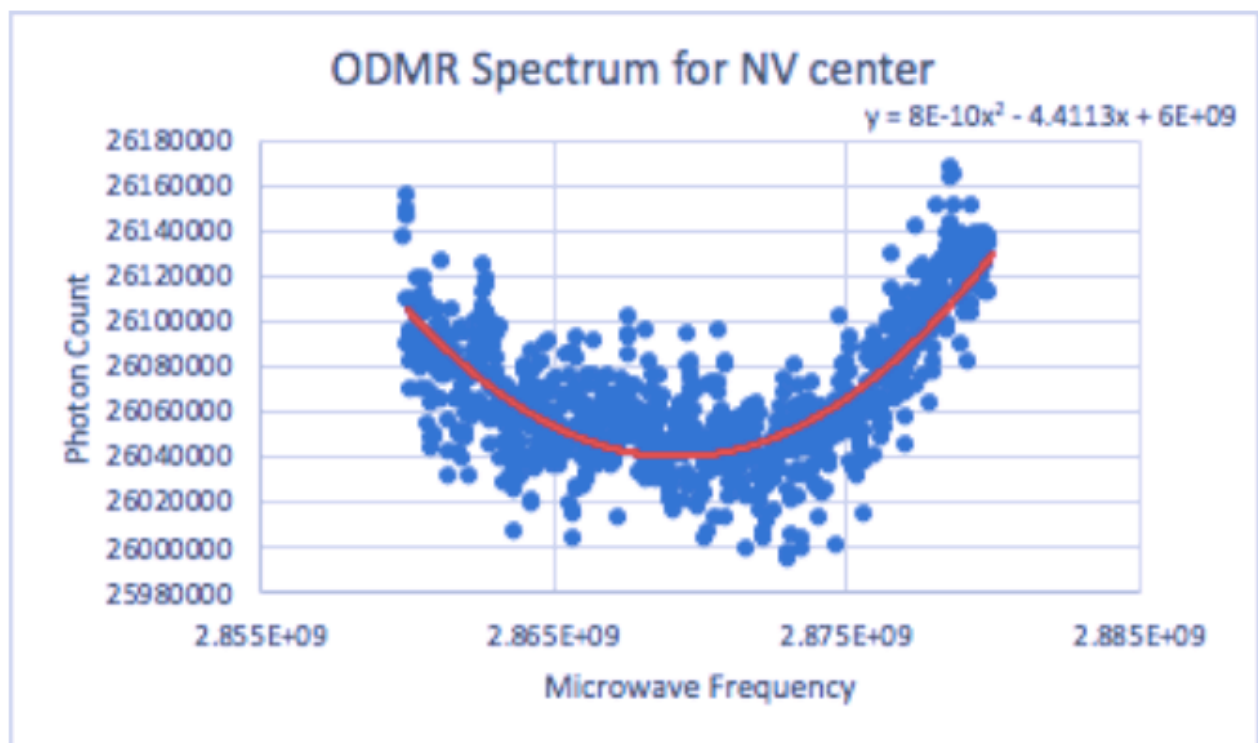


Figure 2.1 – Lucas Wright ‘21’s ODMR spectrum, with a line fit to the data. Reprinted from reference [6].

Over summer 2023 in Oregon, Kai Haesslein, like Wright, produced a frequency vs. luminescence spectrum showing evidence of a changing electron spin state. On top of this, he also produced spectra while magnetic fields were present. To produce these spectra, he used a GCPW, a microwave amplifier, a 532nm laser, an APD, other photon counting equipment, and a software lock-in amplifier.

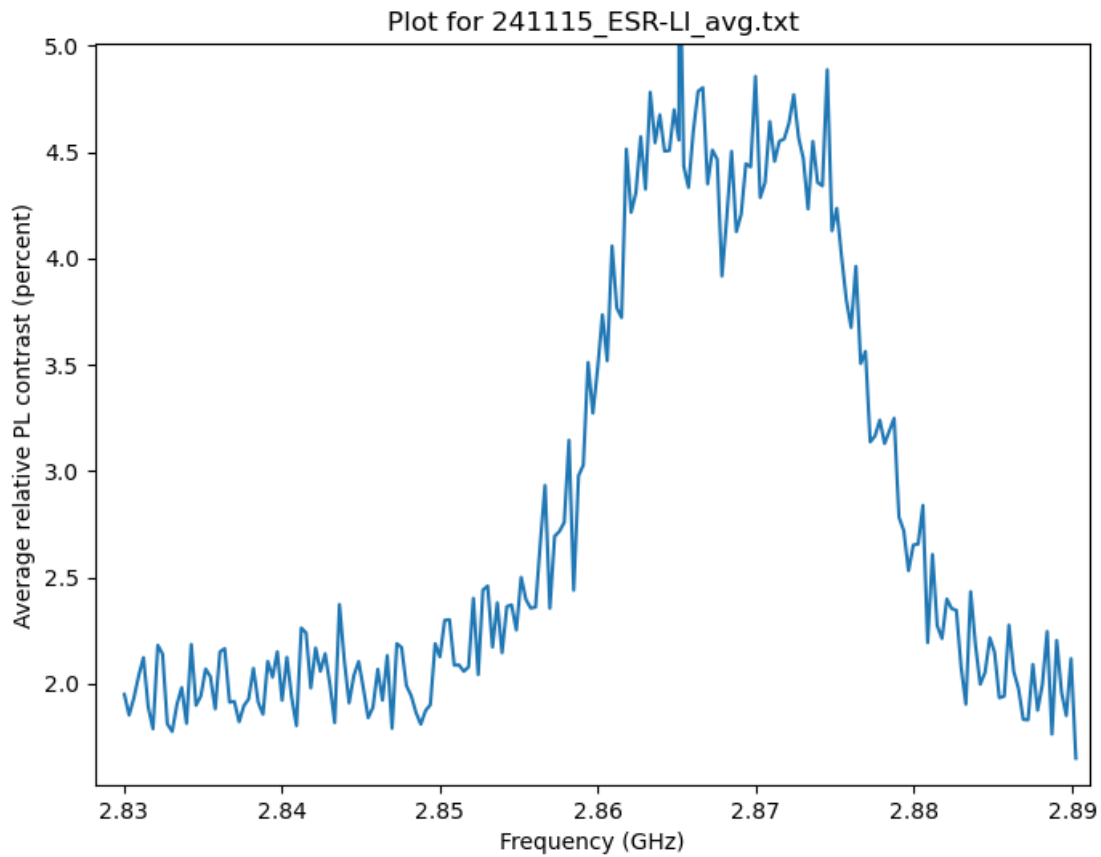


Figure 2.2 – Kai Haesslein ‘24’s ODMR spectrum taken in the presence of a magnetic field, indicated by the double peaks. Data from reference [7].

Although Haesslein has made considerable progress towards the goal of this thesis, the apparatus used in Oregon is not available at Hamilton, and a new system to use NV center diamonds for magnetometry must be engineered using available resources.

3 – Apparatus

3.1 – Photoluminescence detection system

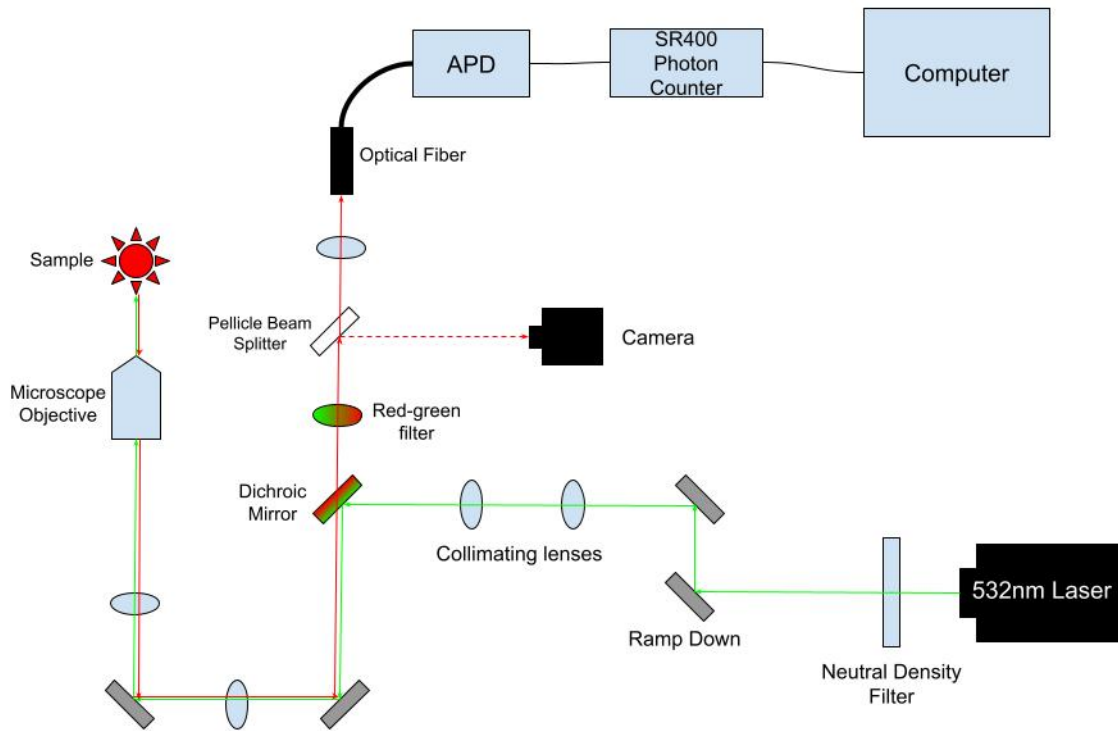


Figure 3.1 – Schematic for confocal microscope setup used by Wright to detect photoluminescence of NV center diamonds.

A confocal microscope was used to detect the NV center diamonds' photoluminescence, and this section will describe the optical path and setup used. Firstly, a 532nm laser beam is first sent through a neutral density filter to attenuate its strength. After the filter, two mirrors lower the beam's height to a Nikon Ti-E Inverted Microscope's optical system's opening. Then, two lenses collimate the beam and send it to a dichroic mirror, which reflects the green laser towards two more lenses that focus the laser into a Nikon Ti-E Inverted Microscope, where the microscope's objective focuses the laser onto our sample.

Once the laser hits the sample, it may cause the NV center to fluoresce by the mechanisms explained in section 1.1. The fluorescent light travels back through the same path that the laser took into the microscope, but it is not reflected by the dichroic mirror due to its longer wavelength. After passing through the dichroic mirror, it passes through a red-green filter that blocks out excess green light. Afterwards, a pellicle beam splitter sends an 8% sample of the beam to a camera, and the remaining 92% passes through where it is focused by a lens into an optical fiber leading to an Avalanche Photodiode (APD). The APD outputs electrical pulses for every photon detected, and these electrical pulses are sent to a SR400 photon counter which then tells a computer the photon count. This is the same confocal microscope apparatus used by Wright to obtain the ODMR spectrum in figure 2.1. A switch was later added in between the APD and SR400 counter for lock-in chopping (see section 3.3).

The optical path was almost perfectly aligned when I started my thesis in the fall and only needed fine tuning with the tilting knobs and translation stages on mirrors and lenses. A cylindrical alignment tool with iris diaphragms on both ends of the cylinder was used to fully align the optical path at the objective. The alignment tool fit onto the microscope's objective carousel, and once the mirrors and lenses were adjusted such that laser was going through the tool with both irises almost completely closed, the laser was aligned and would go in a straight line through the microscope's objective. Another spot in need of re-alignment was the optical fiber; alignment was tested for by disconnecting the fiber from the APD, shining a laser through the disconnected end while the 532nm laser was also on, and then aligning the two laser's paths.

Additionally, the red-green filter is not fixed in place and is removable, and the microscope's stage can also be illuminated by a Sola Light Engine, whose light can be filtered through a quad-band filter and a green filter mounted inside the microscope. The camera's

exposure time and red gain can be adjusted as well. These features were used for aligning the laser with the diamond sample, but not for detecting photoluminescence. Removing the red-green filter allowed some of the excess green light through to the camera, so the exact location where the laser is focused could be seen. The Sola Light Engine combined with its filters illuminated the diamond so we could find it by eye and put it roughly in the center of the eyepiece's field of view, which was within the camera's field of view. Turning up the camera's exposure time and red gain made the diamond easier to spot. Then, moving the stage to put the diamond into the laser's focal position while looking at the camera feed aligned the system for photoluminescence detection.

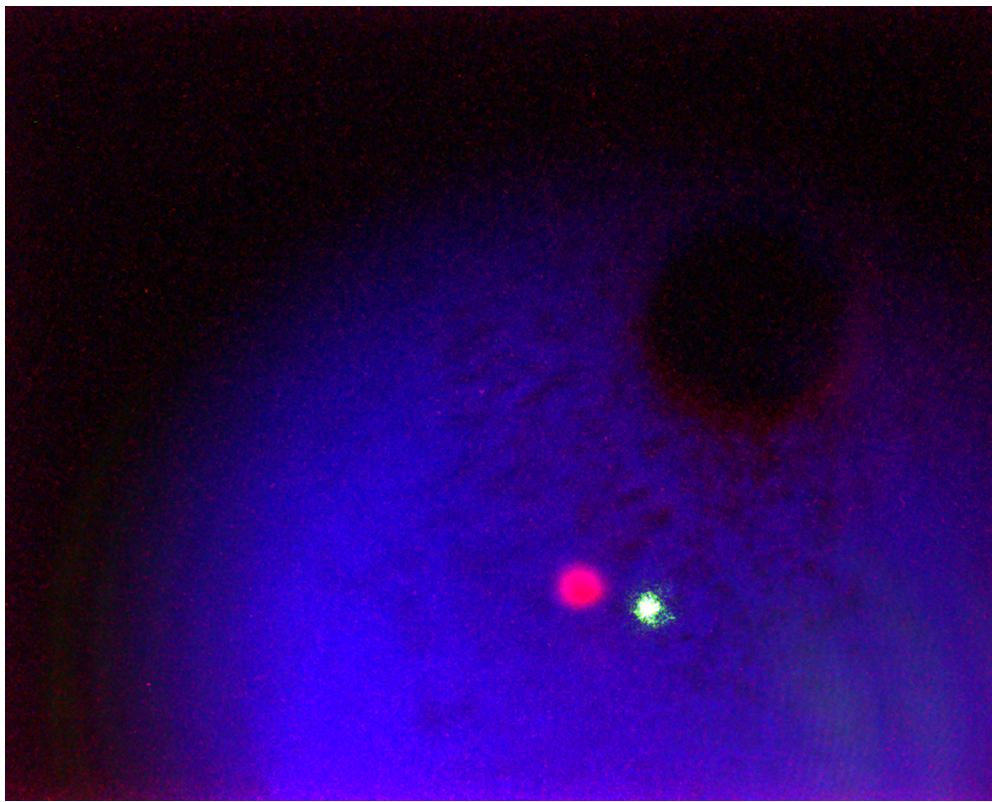


Fig 3.2a – Sample illuminated by Sola Light Engine with the quad-band filter. The green dot is the laser, and the red dot is the diamond sample. The camera's red gain was turned up to the max and the exposure time was 662.82 ms. The laser and sample are unaligned.

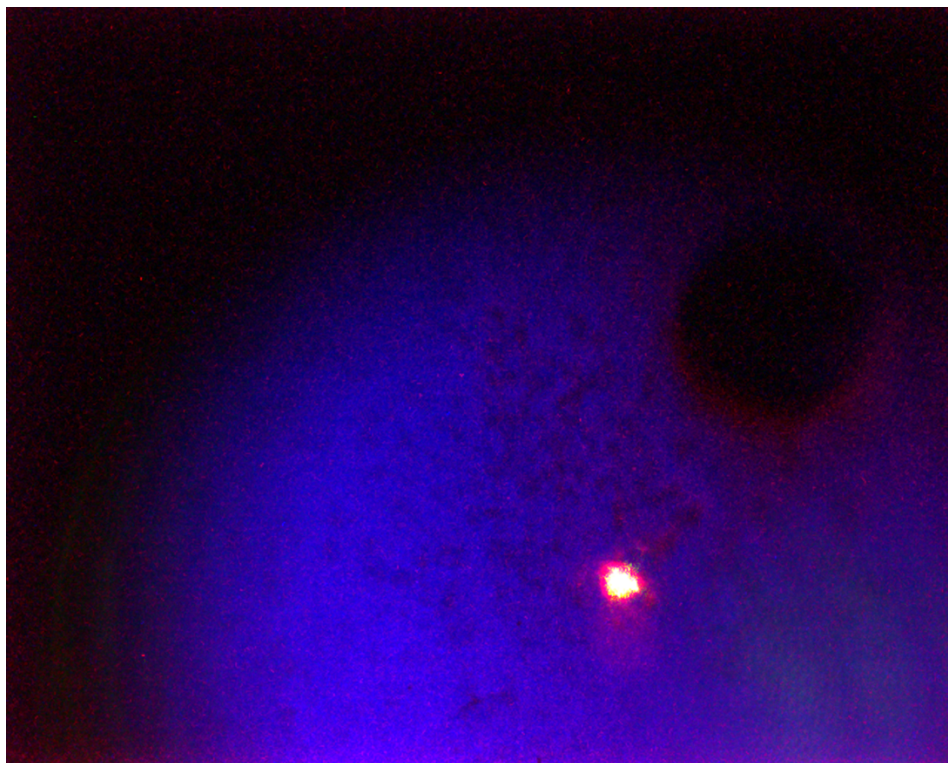


Figure 3.2b – The diamond sample was aligned with the laser by moving the microscope stage. The Sola Light Engine with the quad band filter was still illuminating the sample, and the camera settings were the same as in figure 3.2a.



Figure 3.2c – The Sola Light engine was turned off, and the quad band filter removed. The red-green filter was put in place in the optical path, and the camera's settings were the same as in figures 3.2a and 3.2b. The confocal microscope is now ready to collect photoluminescence data.

3.2 – Antenna

As mentioned in the background section, we used a Grounded Coplanar Waveguide (GCPW) antenna to create the necessary microwave radiation to excite the electrons in our NV centers.

Using an online calculator built upon the equations in section 1.2 [10], we designed our antenna with dimensions $a = 2.85\text{mm}$ and $b = 5.85\text{mm}$. The dimensions h and t were pre-determined by our 1.5mm PCB. These dimensions gave us a characteristic impedance of $50\ \Omega$, allowed us to fit it with an edge-mount SMA connector, and gave us two 1.5mm wide traces to place the diamonds.

The GCPW was designed in Fusion360 and cut from a 1.5mm copper printed circuit board (PCB) in a CNC mill. It was milled using a 0.8mm drill bit and held in place in the CNC mill atop a wooden plank using double sided tape designed specifically for CNC milling and manufacturing.



Fig 3.3 – GCPW in the process of CNC milling.

The first GCPW design was 10mm long (l in figure 3.3), and the lateral ground traces (w in figure 3.3) were 1mm wide. When handling this GCPW, it was quite small and kept falling through the hole in the custom microscope tray (see section 3.3 for tray). Since the GCPW's overall length and the width of its lateral ground strips did not impact its characteristic impedance, the next GCPW design was 20mm long and had 4mm wide lateral ground strips and this was big enough to not fall through a reasonably sized optical viewing hole in the custom microscope tray.

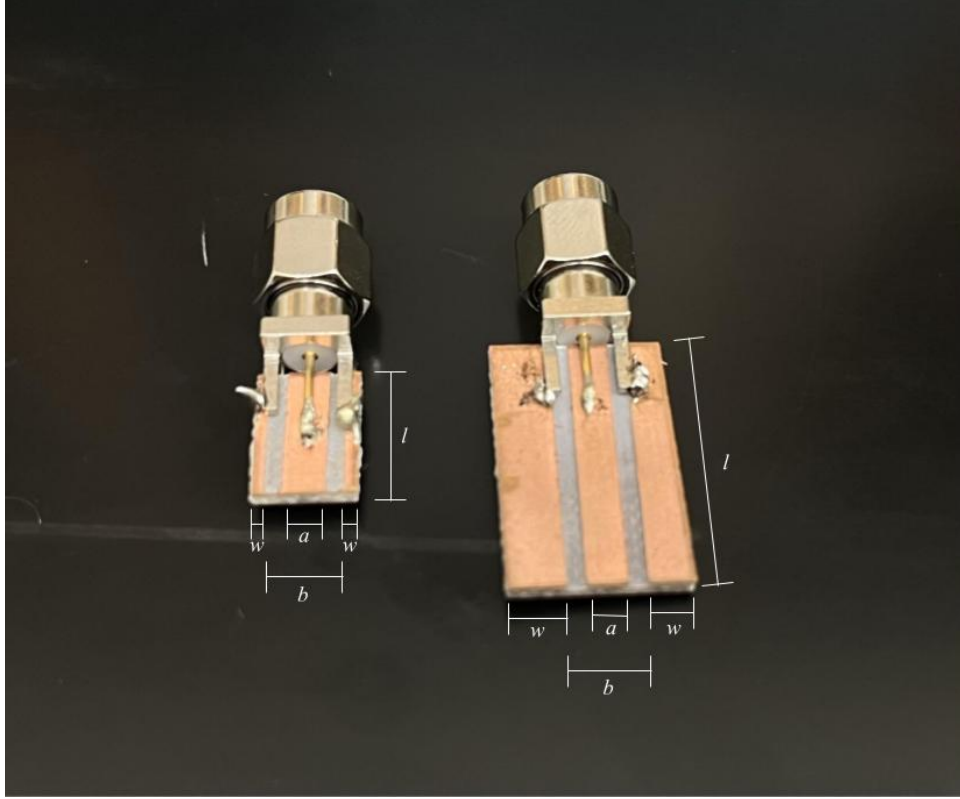


Figure 3.4 – The first GCPW design on the left, and the second on the right. The dimensions l and w were changed so the antenna wouldn't fall through the optical viewing hole, but a and b were unchanged to keep the characteristic impedance at $50\ \Omega$.

The antenna was then connected to a SG380 Series RF Signal Generator by an edge-mount SMA connector, an SMA cable, and an N-type to SMA adapter. The SG380's signal is converted to microwave radiation by the GCPW.

After unsuccessful ODMR data collection (see sections 4.1 and 4.2), a Mini-Circuits ZHL-15W-422-S+ RF signal amplifier was added. The SG380 output went into the amplifier, and the amplifier's output went to the GCPW. The amplifier greatly increased the microwave signal strength from the GCPW. Before adding the amplifier, even when the SG380 was outputting signal at its maximum amplitude, 16.5 dBm, the GCPW's signal strength was -44.8 dBm. When the amplifier was connected and the SG380 was outputting the amplifier's maximum input signal amplitude, 7 dBm, the GCPW's signal strength was -18.7 dBm. Since power in dBm is given by the equation $P_{dBm} = 10 \log_{10} \left(\frac{P_{mW}}{1\ mW} \right)$, where P_{dBm} is the power in

dBm and P_{mW} is the power in watts, the 26.1 dBm increase in the signal meant that the radiation became over 407 times more powerful.

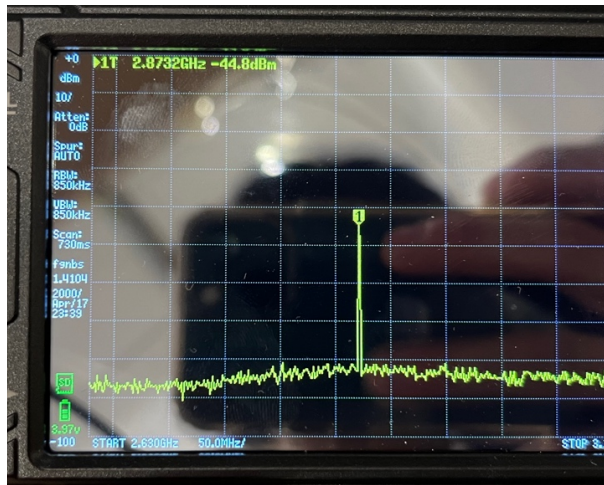


Figure 3.5a

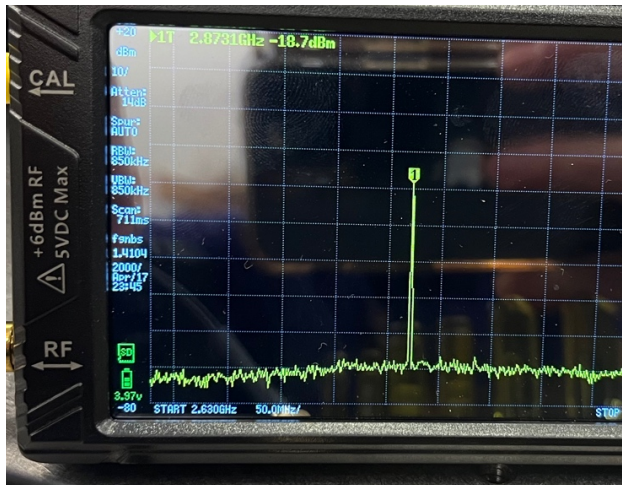


Figure 3.5b

Figure 3.5a, 3.5b – Spectrum analyzer outputs of antenna signal before and after amplification. Figure 3.5a is without the amplifier and the SG380 outputting 16.5 dBm; the antenna signal peaks at -44.8 dBm. Figure 3.5b is with the amplifier and the SG380 outputting 7 dBm; the antenna signal peaks at -18.7 dBm.

3.3 – Focusing Microscope onto Antenna

The Nikon Ti-E Inverted Microscope's stage was previously only able to hold 3" x 1" slides within the microscope's focal length, and thus the stage had to be modified to hold the GCPW at a position where the microscope can focus on its traces, where the diamonds are. The stage attachment for glass slides was removed and was replaced by a 3D printed tray with a lowered platform to hold the antenna in focus.

The first tray design had a platform lowered 5mm, arms specifically positioned to rest upon the microscope stage, and the platform's dimensions were decided by the arms' intersection points. A mill was used to manually drill a hole through the platform for the objective to look through. Although this successfully could hold the GCPW within the microscope's focus, drilling the optical viewing hole deformed the 3D printing filament around it, preventing the

GCPW from lying flat, and stray filament partially obscured parts of the GCPW when viewed through the microscope.

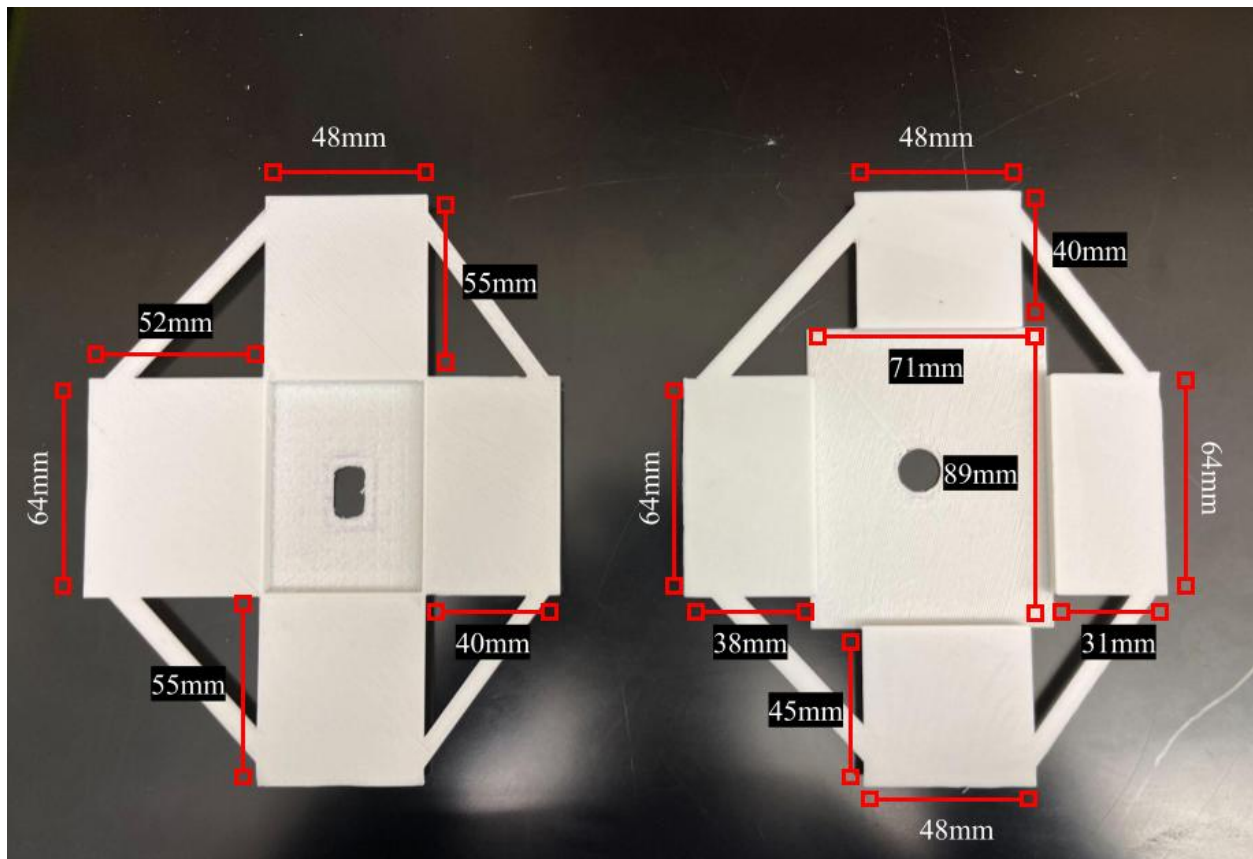


Figure 3.6 – First tray design on the left, and second tray design on the right with dimensions shown. Second tray design has a cleaner, circular hole, a larger platform to place samples on.

The newer tray design had a larger platform, openings on the sides of the stage for the antenna's signal cable to feed through, and an opening in the bottom for the objective to look through already designed in the 3D model, avoiding the need to drill. The tray was held still on the stage by tape, and another piece of tape positioned the GCPW above the tray's hole.

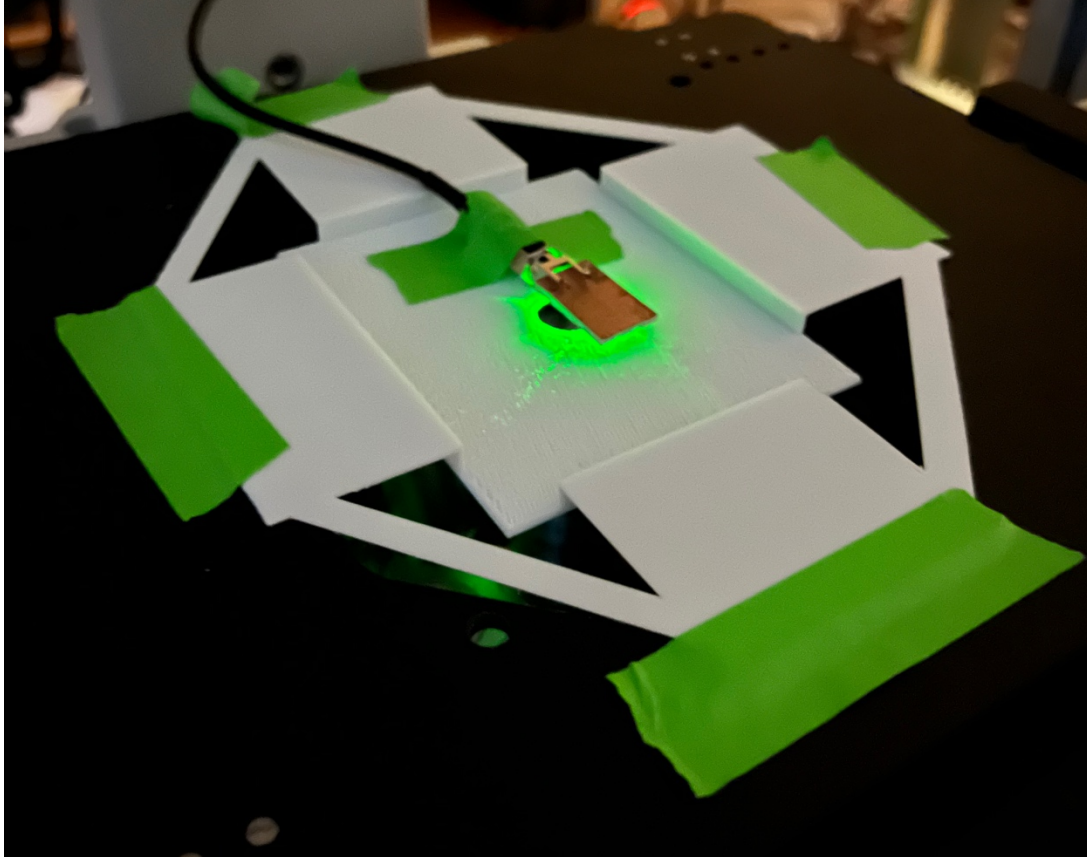


Fig 3.7 – GCPW in the newer custom 3D printed tray.

3.4 – Lock-in Chopping

A python program written by Wright (see appendix A) controlled the signal generator and read the SR400's photon counts simultaneously – it did a random sweep of the frequencies within a given range and collected the photon count over 1 second at each of those frequencies. Due to constant photobleaching, which is when a fluorescent substance permanently loses its fluorescence, this method no longer worked for data collection (see 4.1), and Lock-in Chopping was implemented as a corrective measure. Photobleaching caused the number of fluorescent NV centers to decrease in the sample over time, and thus constantly decreased the fluorescence of the diamond sample over time, even without any microwave radiation present.

To implement Lock-in Chopping, a Mini-Circuits ZYSWA-2-50DR+ switch was used in conjunction with the SG380's pulse modulation function, both of the SR400's built-in photon

counters were used, and Wright's code was adapted to collect data using Lock-in Chopping.

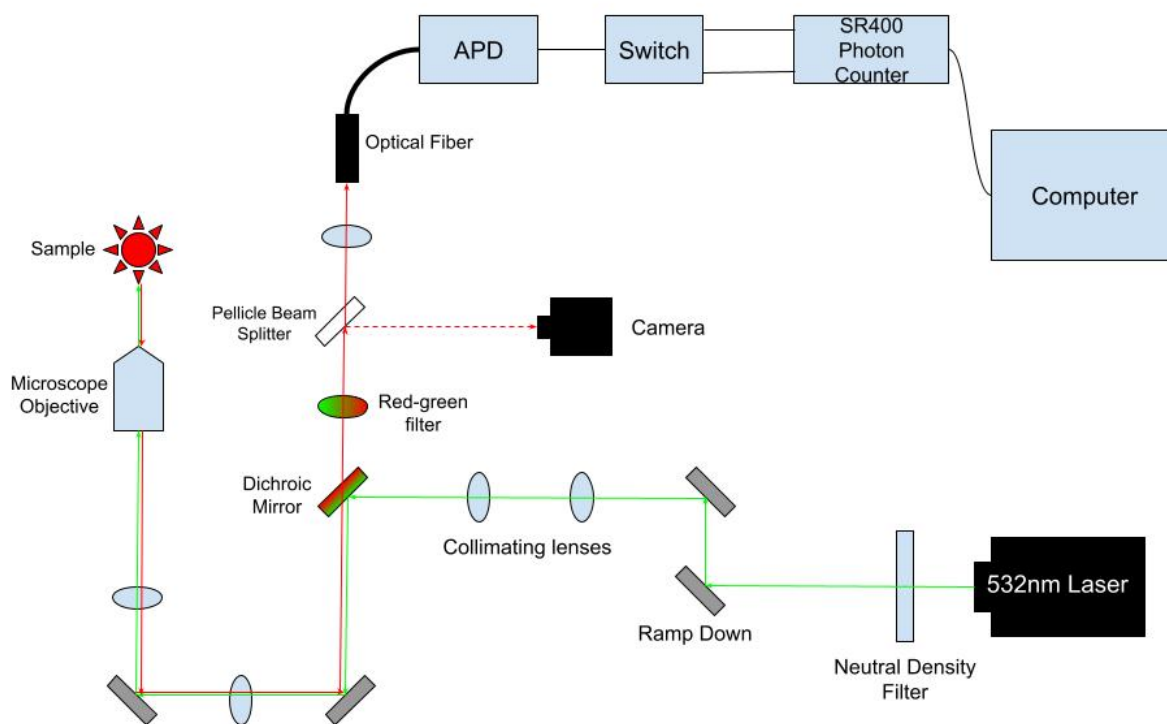


Figure 3.8 – Overall schematic for confocal microscope setup used to detect photoluminescence of NV center diamonds with Lock-in Chopping. More detail on the switch's setup in fig. 3.9.

The Mini-Circuits ZYSWA-2-50DR+ switch takes an input signal, and sends it to one of its two outputs, depending on the state of its TTL input – when the TTL is low, the input signal is sent to output RF1, and when high, the input signal goes to RF2.

The SG380 Signal Generator's pulse modulation function envelopes its programmed signal within a square wave. The SG380's modulation output on the back outputs the signal's envelope, which is a square wave when the pulse modulation function is on.

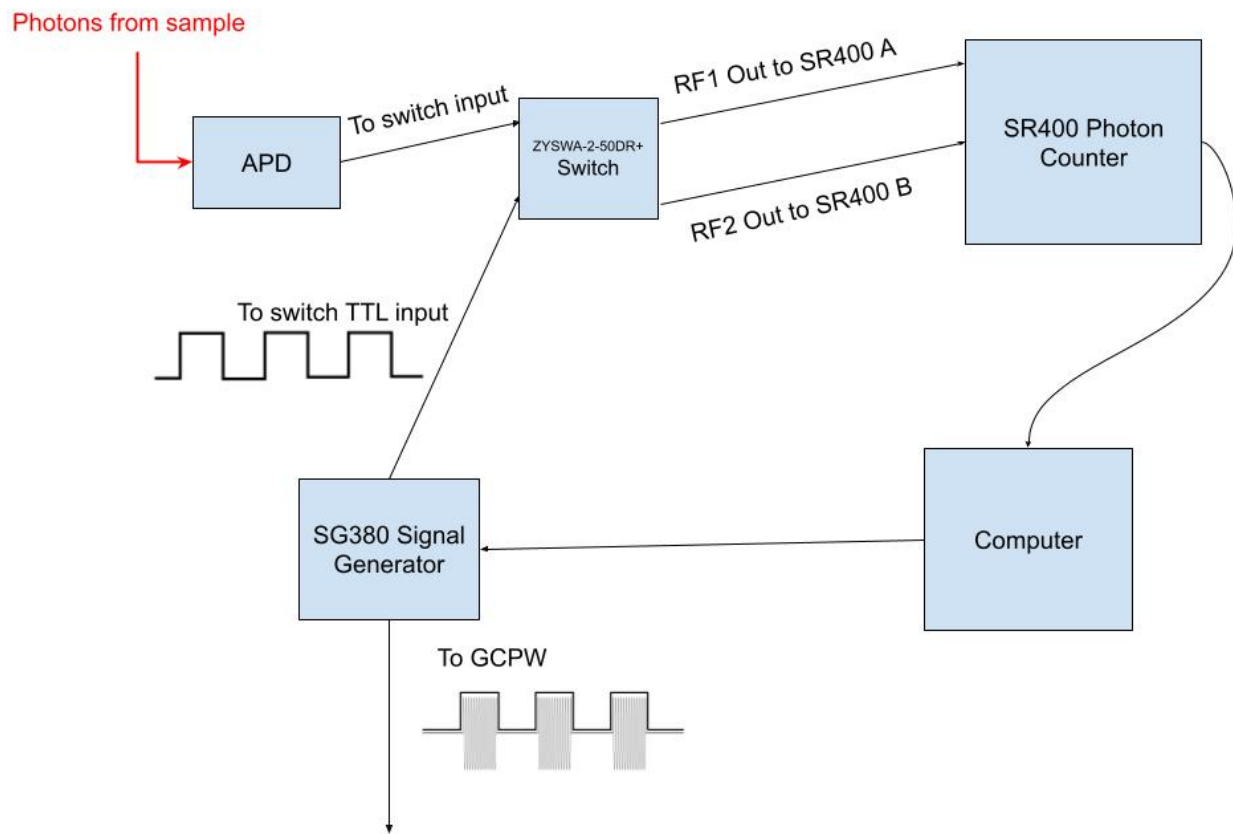


Figure 3.9 – Schematic for implementation of Lock-in Chopping.

By connecting the APD signal to the switch's input and the SG380 modulation output to the switch's TTL input while using the pulse modulation function on our microwave signal, we separated between control measurements and active measurements. Output RF1's signal gave photon counts while the microwave was off, and RF2's signal gave photon counts while the diamond was exposed to microwave radiation. RF1 was sent to the SR400's photon counter A, while RF2 was sent to the SR400's photon counter B.

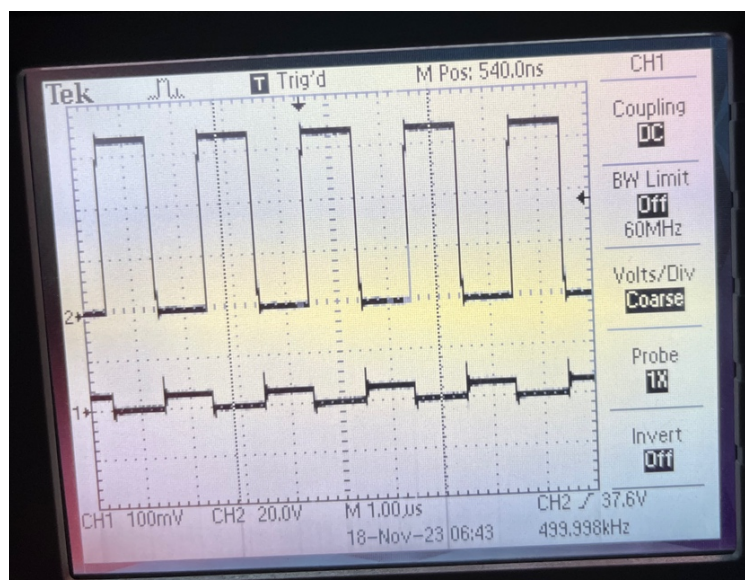


Figure 3.10a

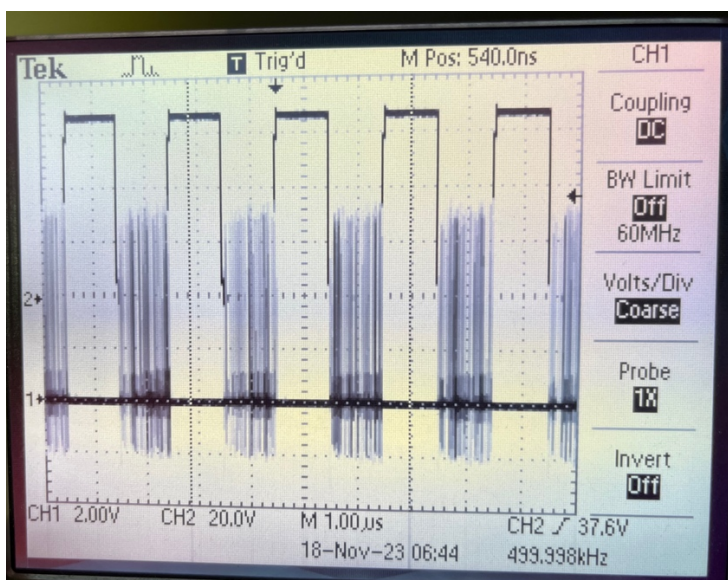


Figure 3.10b

Figure 3.10a & 3.10b – Signal from APD being modulated by switch. Figure 3.10a is when there is darkness (no photons), and figure 3.10b is with a flashlight pointing at the optical fiber. The bottom signal is from switch output RF1, and the top is the modulation signal from the SG380.

Wright's code originally swept through a range of frequencies in a random order while collecting photon counts at each of those frequencies. Because of the constant decrease in luminosity of the diamond NV center caused by photobleaching, Wright's code was edited and adapted (see appendix B) to take photon counts in both A and B over a second and find the difference between those counts for each frequency in the program's given range. The code could be toggled to sweep through the frequency range randomly or in sequential order. A random sweep could mitigate systematic error due to photobleaching or drift, since data could be taken throughout the frequency range before photobleaching or drift begins to affect the quality of the data, and thus systematic error would not heavily affect only a concentrated part of the spectrum and instead mildly impact the overall spectrum.

The A and B photon counts were taken over the same second, and by setting the pulse modulation function's duty cycle to 50%, half of each second-long measurement's photon counts are control measurements while the other half are while the antenna is emitting a microwave

signal. Since the luminosity will decrease at roughly the same rate throughout this second-long measurement, using the difference between the counts in A and B give us a relative luminosity corresponding to the frequency the GCPW was emitting.

Originally, the new code took a measurement over a second through counter A, took a measurement over the next second through counter B, found the difference between the A and B counts, and then repeated for every frequency in the given range. While this worked, it was much slower than interposing the two measurements within the same second, and the code was revised to interpose the two, greatly saving time, while offering better results; taking interposed measurements through A and B meant they could more effectively cancel out noise since they both measure the same noise over the same second.

Another addition to the code was averaging. The averaging value could be changed to take more or less data at each frequency. For example, if the averaging value was 5, then 5 1-second-long measurements would be taken at every frequency. The code took the differences between the A and B photon counts during each of those 1-second-long measurements, then averaged those differences, and this average difference between A and B value gave us the relative luminosity at each frequency.

4 – ODMR progress

4.1 – Using same measurement method as Wright

Photoluminescence data in this section was collected using the same code (appendix A) and confocal microscope setup (figure 3.1) as Wright, meaning there is no Lock-in Chopping. Attempting to measure a decrease in luminescence of the diamond NV center caused by a changing spin state returned an unintelligible ODMR spectrum of seemingly random data points.

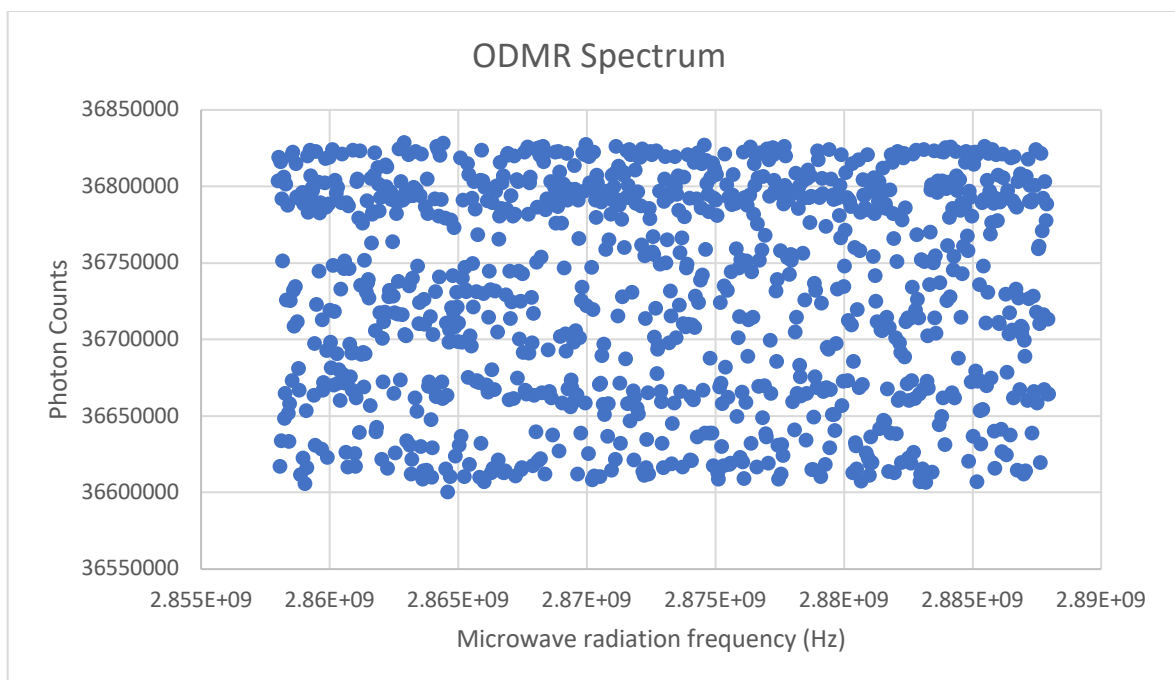


Figure 4.1 – ODMR spectrum taken of diamond NV center using wright’s code. Every data run more or less looks like this, a jumble of points with no discernible pattern.

After multiple runs, I noticed that the photon counts were decreasing every consecutive measurement, regardless of the microwave radiation’s frequency – this was caused by photobleaching. Since Wright’s code swept through its given frequency range in a random order, the photon counts decreasing as time passed meant the collected data ended up completely random.

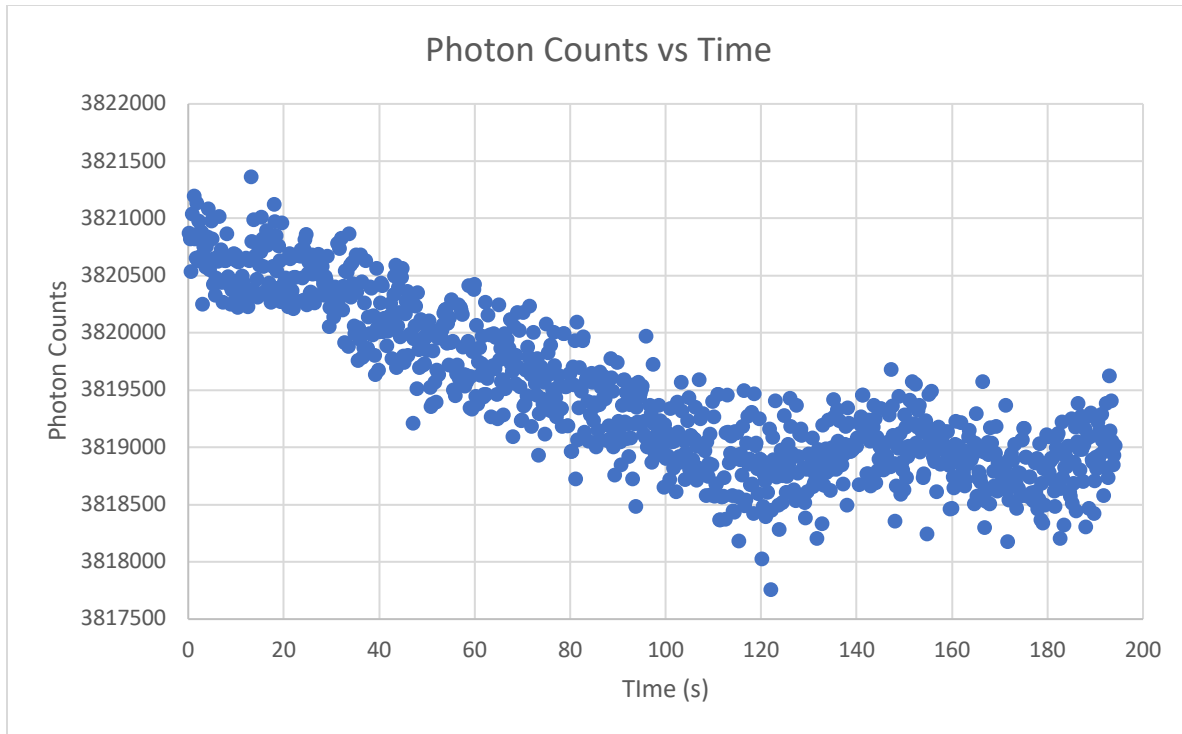


Fig 4.2 – Diamond NV Center photon counts taken over time, without any microwave radiation present. The photon counts steadily decrease without any external changes.

Thus, Lock-in Chopping was applied as a combative measure to take data amidst the steady decrease in fluorescence.

4.2 – Using Lock-in Chopping

Photoluminescence data in this section was taken using the confocal microscope setup, measurement hardware, and Wright's code edited and designed to implement Lock-in Chopping (see section 3.4). Firstly, the measurements were taken by sweeping through the frequencies in sequential order, and then in random order.

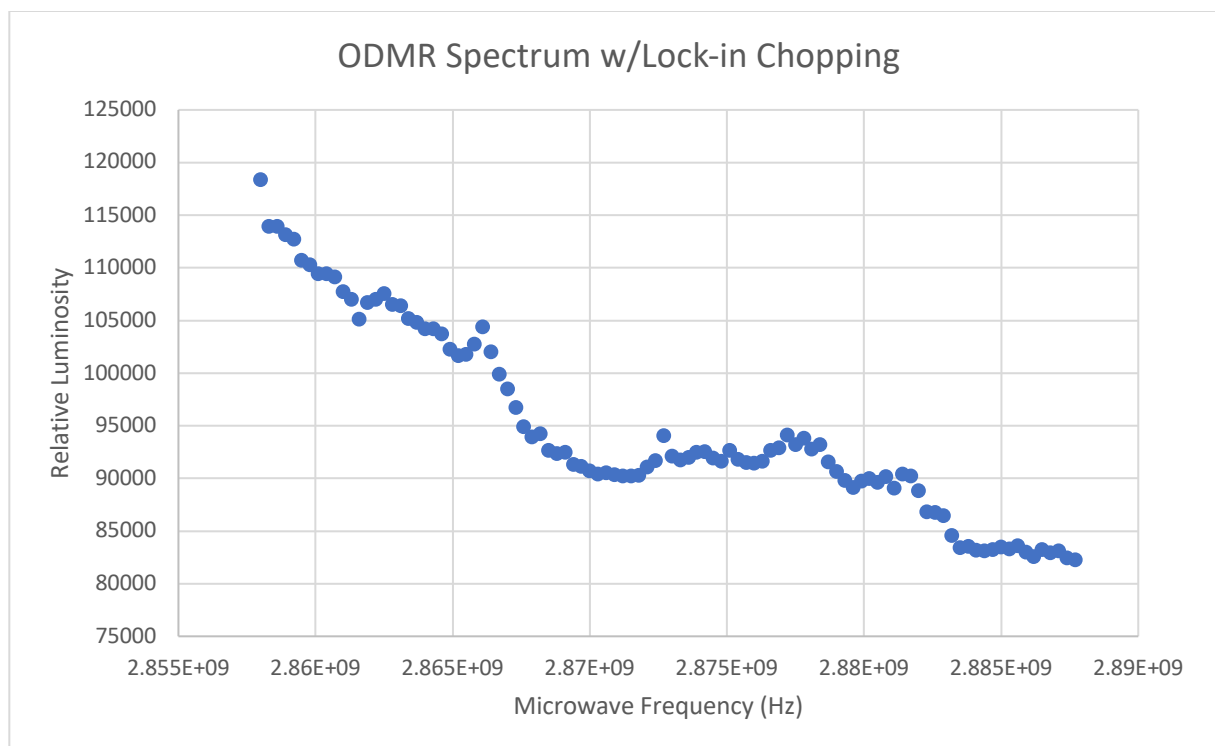


Figure 4.3a – ODMR Spectrum taken with Lock-in Chopping. Frequencies were swept through in sequential order. The dip at 2.87 GHz shows promise, but it is not enough to say it really is a dip and similar data was never reproduced or improved upon.

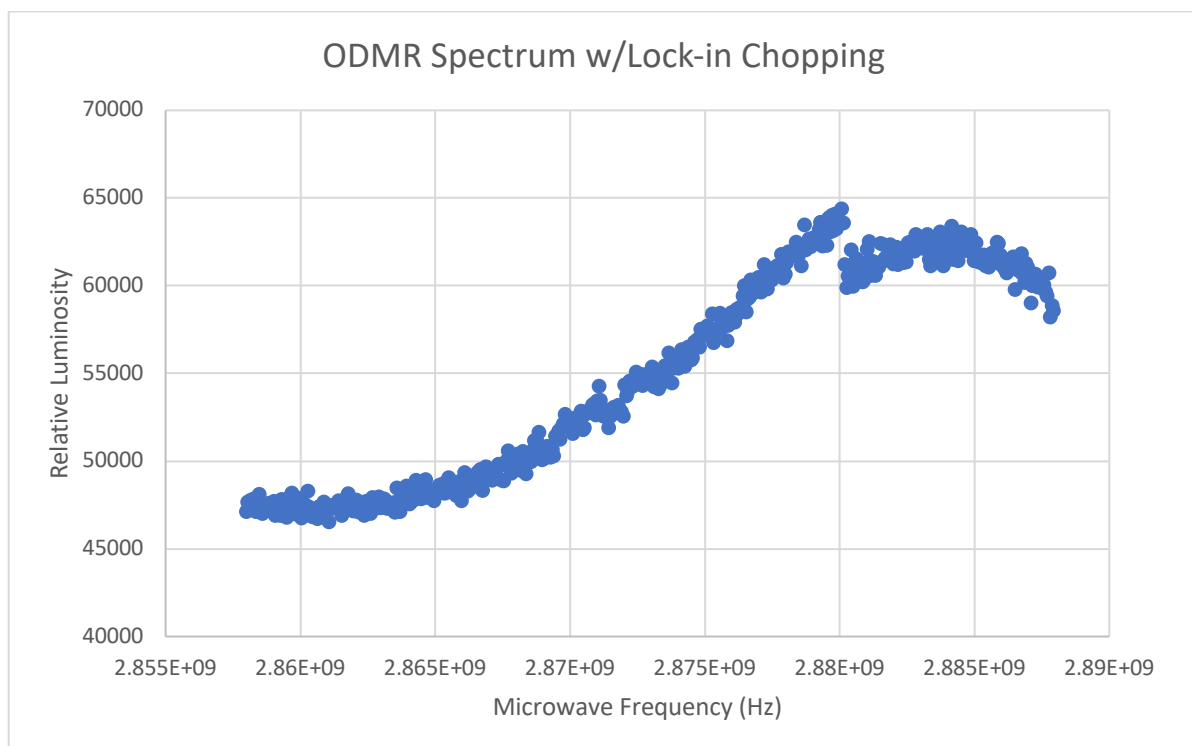


Figure 4.3b - ODMR Spectrum taken with Lock-in Chopping. Frequencies were swept through in sequential order. The data shows a pattern, but it is not a dip as expected. Again, further data collection never reproduced a similar looking plot.

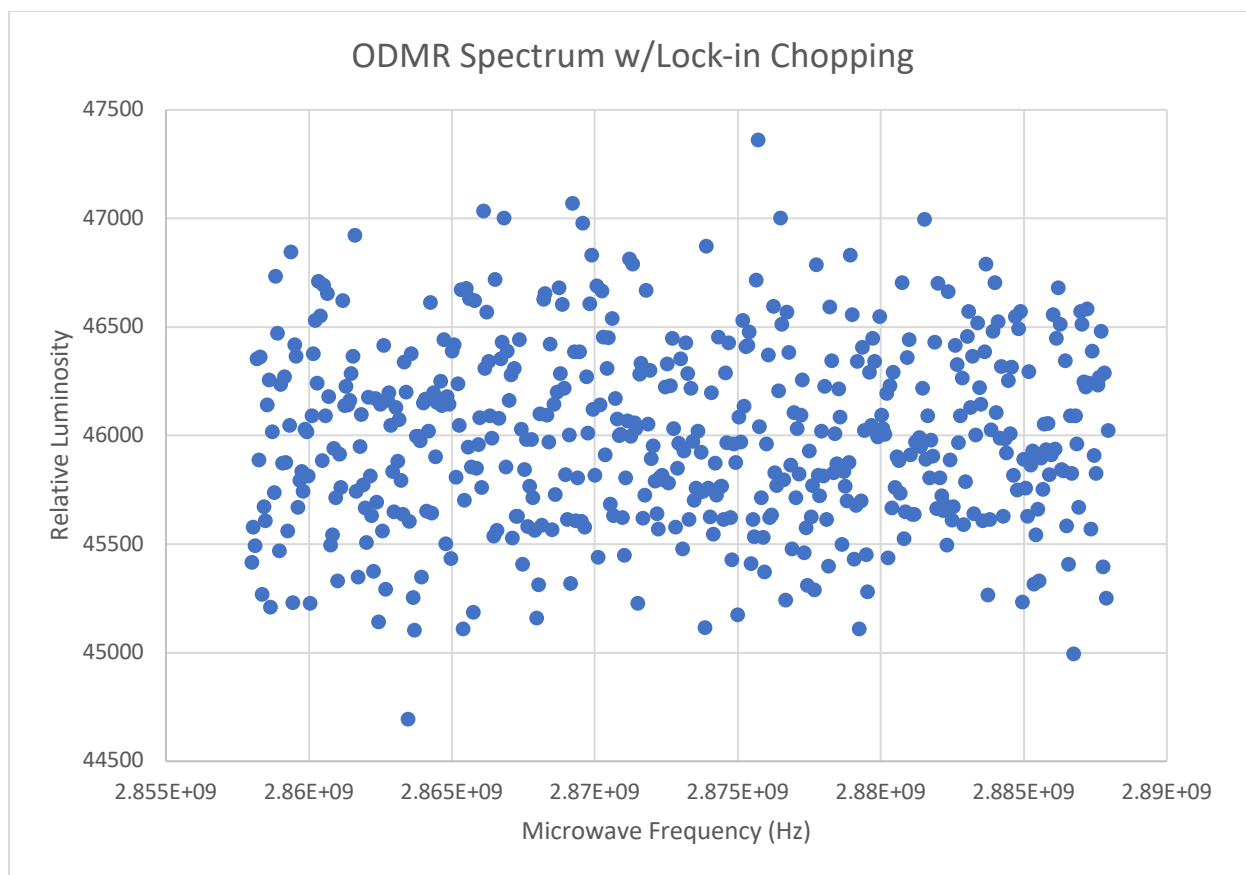


Figure 4.3c - ODMR Spectrum taken with Lock-in Chopping. Frequencies were swept through in random order. Note that most spectra taken look like 4.3c even when sweeping through sequentially – 4.3a and 4.3b were just notable examples.

Unfortunately, deploying lock-in chopping was not an immediate success. For an unknown reason, the data was still skewed, didn't show evidence of a changing spin state, or was completely random. Both the sequential and random sweeps through the given frequencies yielded random-looking spectra, but the least random looking spectra came from sequential sweeps (fig. 4.3a and 4.3b). This may have been caused by the sample drifting out of alignment due to the 3D printed tray thermally expanding and contracting.

As an effort to ameliorate these seemingly erratic measurements, the code was adjusted to use the saved time from interposing the counter A and B measurements to take more measurements at each frequency. Numerous 1-second-long measurements could be taken at each

frequency, and the difference between A and B for each of those measurements were averaged, and this mean of the differences was used as the relative luminosity for each frequency.

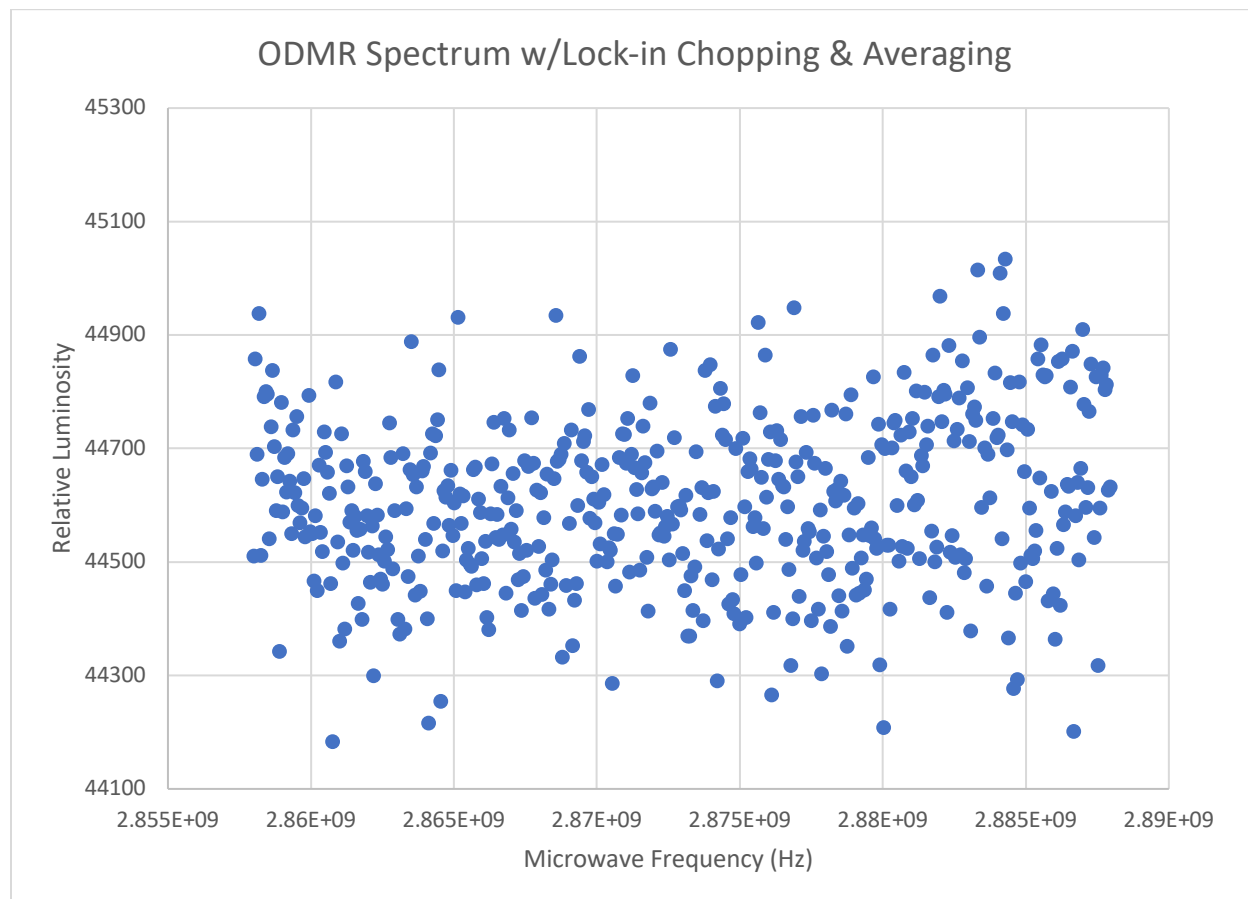


Figure 4.4 – ODMR Spectrum taken with Lock-in Chopping & averaging over 5 measurements at each frequency. The frequency was swept through in random order.

Adding averaging was unsuccessful as well, as the data still came out looking random once again. Various spectra taken with averaging ranging from 2-5 measurements per frequency all resemble figure 4.4, regardless of the sweeping method.

4.3 – Adding Amplifier to SG380 Signal

After taking a plethora of ODMR scans without any data showing evidence of a changing spin state, the amplifier mentioned at the end of section 3.2 was added and increased the GCPW's microwave signal strength by over 407 times.

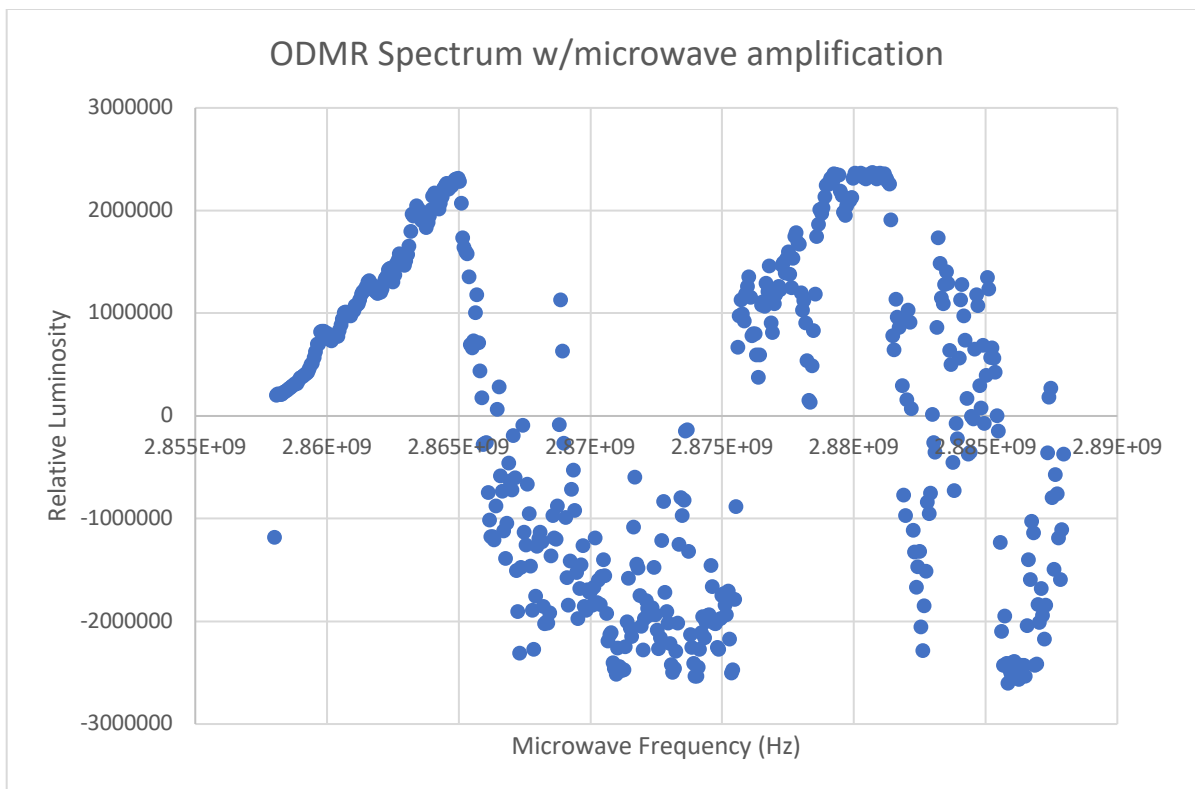


Figure 4.5a – ODMR Spectrum taken with an amplified microwave signal, Lock-in Chopping, no averaging, and a sequential frequency sweep.

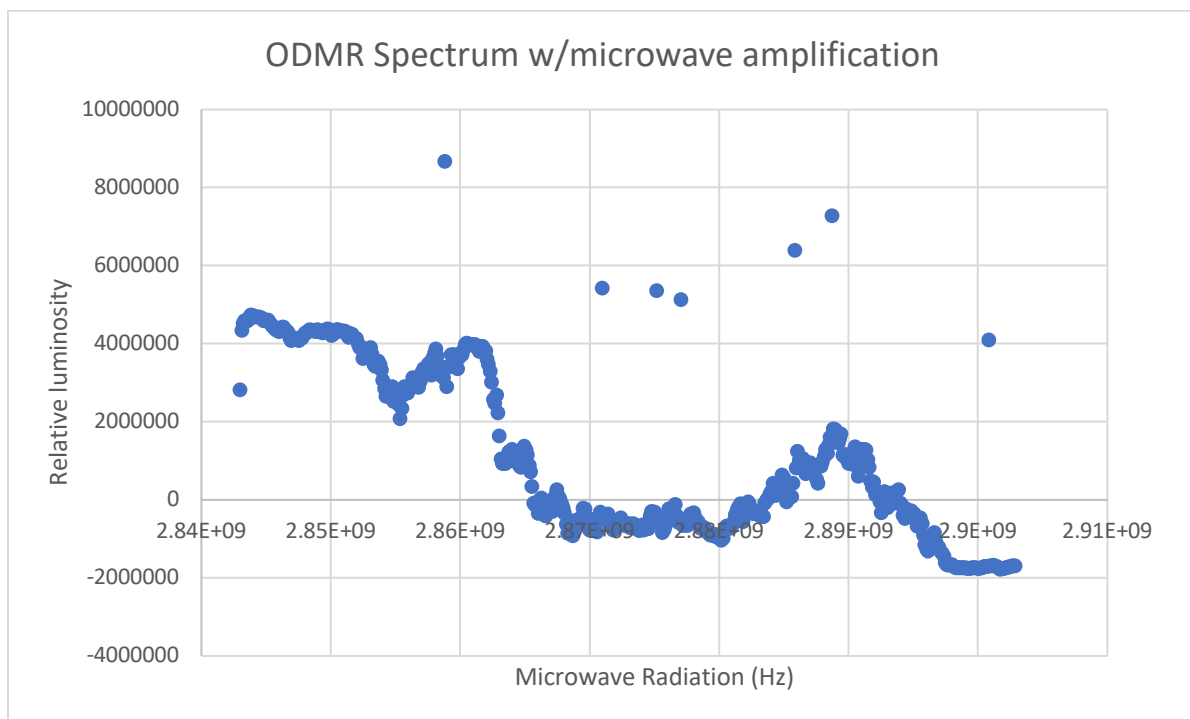


Figure 4.5b - ODMR Spectrum taken with an amplified microwave signal, Lock-in Chopping, averaging over 3 points per frequency, and a random frequency sweep.

The ODMR spectra taken were far from perfect, but they definitely showed evidence of a changing spin state. Every data run was totally different – some still looked like figure 4.4, while others, like figures 4.5a and 4.5b, showed the decrease in luminosity around 2.87 GHz indicating a changing spin state. The data collected was very inconsistent and unpredictable.

These spectra taken without magnetic fields still had lots of room for improvement, but I went ahead and took a couple spectra with a Neodymium magnet on the microscope stage 11.5 cm away from the sample, just to see if I could get close to seeing double peaks.

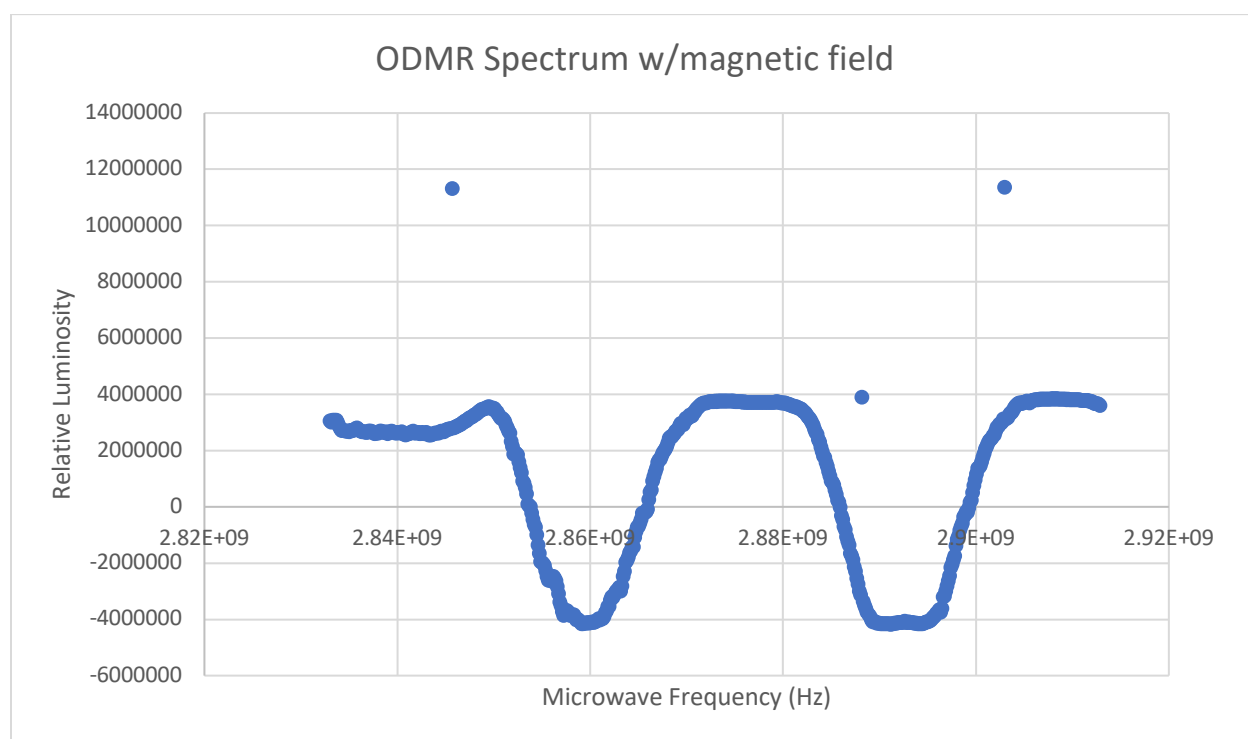


Figure 4.6 – ODMR Spectrum taken in the presence of a magnetic field. This was taken with an amplified microwave signal, Lock-in Chopping, averaging over 2 points per frequency, and a sequential frequency sweep. Undoubtedly the clearest spectrum taken.

A very clear ODMR spectrum in the presence of a magnetic field was obtained using Lock-in Chopping, averaging over two measurements at each frequency, and a sequential frequency sweep. Every other data run in the presence of magnets did not show any evidence of a changing spin state, let alone double peaks. This level of clarity had never been achieved before and also was unfortunately unable to be reproduced.

5 – Results/Conclusion

By fitting gaussian peaks to the data from figure 4.6, the magnetic field present was measured using the theory in section 1.1.

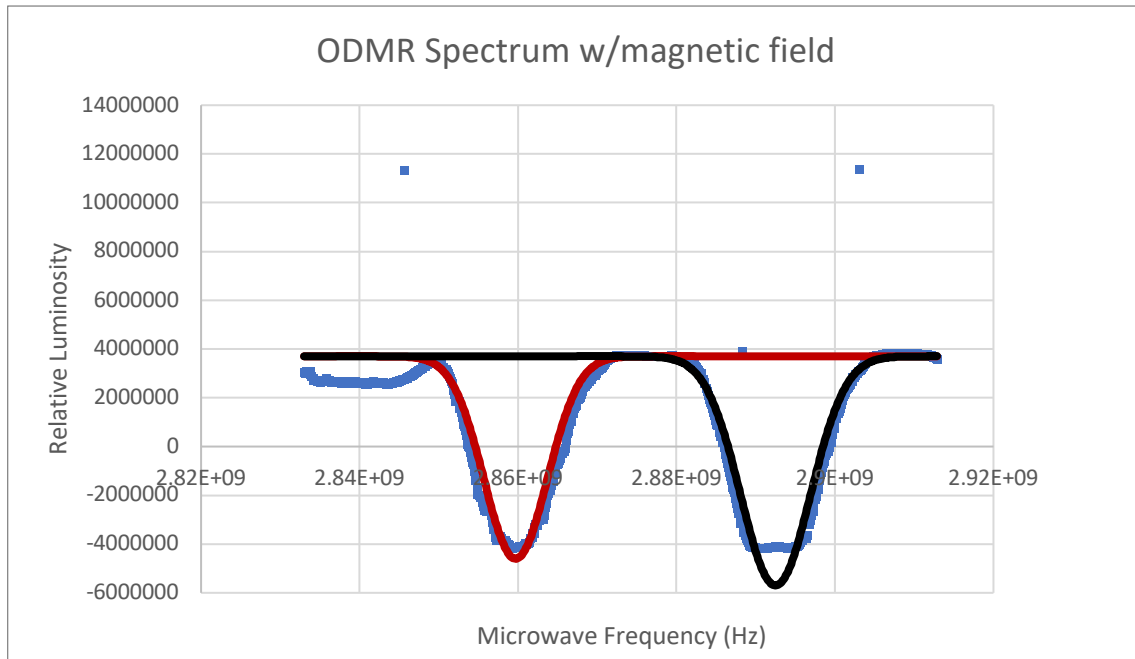


Fig 5.1 – Data from figure 4.6 with gaussian peaks fitted. The red and black lines are the fitted peaks, and the blue is the data from 4.6.

Gaussian peaks were fitted over the data from figure 4.6 by eye. This method revealed that the centers of each peak were at 2.8597 GHz, and 2.89256 GHz. Using $E = 2\gamma B_z$ and the gyromagnetic ratio $\gamma = 2\pi * 28 \text{ GHz/T}$, we get that the magnetic field is $B_z = 9.3380 * 10^{-5} \text{ T} = 0.93380 \text{ G}$.

Only one magnetic field measurement was made due to the inconsistencies of data collection. Furthermore, the potential error on this data cannot be truly known because the peak fitting method was so rudimentary. Using solver on excel to fit double peaks lead to many issues, and even caused my computer to crash.

Presently, the entire system possesses the capability to measure magnetic fields through ODMR spectra; however, its precision remains uncertain, and it exhibits significant

inconsistency. While it demonstrated promise with the distinctly clear ODMR spectrum in Figure 4.6, it struggles to consistently replicate such sharply defined spectra.

6 – Future Work

There is lots of room for improvement with the current system. Only one clear ODMR spectrum was obtained, and the inconsistency of the system is its biggest issue.

An improvement that could completely change the quality of data taken is by redesigning the 3D printed tray that the GCPW sits upon in the microscope stage, and/or by changing the design of the GCPW itself to allow for the use of more powerful objectives. Using more powerful objectives could focus the laser into a smaller point, which could increase the diamond's fluorescence, increasing the photon count signal. Currently, any microscope objective more powerful than 10x magnification cannot be used because it touches the bottom of the tray, pushing it upwards, making the objective unable to get close enough to the sample to get it in focus.



Figure 6.1a



Figure 6.1b

Figure 6.1a, 6.1b – Photos taken from underneath the microscope stage. In fig. 6.1a, the 10x objective is in use, and in 6.1b the 20x objective is in use. The 20x objective touches the underside of the tray making it unable to get closer to focus.

A possible design change could be widening the tray's optical viewing hole and increasing the size of the GCPW without changing its impedance to ensure it doesn't fall through the enlarged hole. This way, the heightened conical tip of the more powerful objectives wouldn't touch the tray while getting closer to the sample in the GCPW.

An alternate design change would be to change the GCPW design or make precisely shaped divots or openings in the tray so the GCPW can lie flat directly upon the surface of the tray's platform. Currently, the edge-mount SMA connector soldered to the GCPW protrudes such that the GCPW is hovering slightly above the tray's platform surface, necessitating microscope objectives to move further upwards before the sample is in focus. A hole or opening in the tray's platform big enough to let the protruding parts of the connector through such that the antenna lies flat on the platform would bring the antenna's resting point further downwards and remedy this problem. Alternatively, the GCPW could be designed to fit with an SMA connector that mounts on top of the GCPW's ground plane and the tray wouldn't need modifications in order to get the GCPW to lie flat. Either of these alterations could allow the use of more powerful objectives without their tips touching the tray. Furthermore, this would increase the stability of the GCPW as it rests upon the tray, and the GCPW would not be jolted out of position by small accidental contacts with the signal cables.

Another improvement to the overall system would be to make the custom tray out of a material less prone to thermal expansion and contraction. The 3D print filament making up the current tray is sensitive to minute temperature changes, and this causes the sample to shift in short time scales (~5 min), slowly moving it out of alignment with the laser, affecting the accuracy of the data collected, especially for longer data runs with more averaging that is supposed to be more accurate.

An alternative option to keep the diamond from drifting out of alignment with the laser would be to design and establish a means to fit the microscope with the currently unused “Nanopositioner” robotic moving stage. Using this robotic stage, a program could be written for the stage to move such that it counteracts the diamond’s drift, holding it in alignment with the laser during data collection.

While the current lock-in chopping setup with the switch works totally fine, the SR400 photon counter has a built in Lock-In/Chopped mode that can do the exact same job as the switch. Although this wouldn’t offer any immediate improvements to the design, it would simplify the photon counting and data collection setup.

Once data collection has reached a point of reliably reproducing clear ODMR spectra, the process of getting the magnetic field out of the spectra can also be refined. While gaussian peak fitting by eye gave a decent approximate fitting, truly minimizing the residuals with Excel’s solver or python code would yield more accurate peak fittings, leading to more accurate magnetic field measurements.

References

- [1] *Diamond NV Center magnetometry*. NIST. (2020, October 6).
<https://www.nist.gov/programs-projects/diamond-nv-center-magnetometry>
- [2] Schirhagl, R., Chang, K., Loretz, M., & Degen, C. L. (2014). Nitrogen-vacancy centers in Diamond: Nanoscale Sensors for Physics and Biology. *Annual Review of Physical Chemistry*, 65(1), 83–105. <https://doi.org/10.1146/annurev-physchem-040513-103659>
- [3] *Magnetometry with single NV centers in diamond*. Magnetometry with single NV Centers in Diamond. (n.d.). https://www.qolah.org/research/nv_v2/nv_v2.html
- [5] Rogers' Advanced Electronics Solutions. (2015). *Microstrip vs. Coplanar Waveguide*. YouTube. Retrieved November 23, 2023, from <https://www.youtube.com/watch?v=SwNb0ndIT5k>.
- [4] Wadell, B. C. (1991). *Transmission Line Design Handbook*. Artech House.
- [6] Wright, L. (2020). *Progress Toward Microfluidic Nanodiamond Sorting* (thesis).
- [7] Haessein, Kai. (2023). *NV Diamond Research in Oregon* (summer research).
- [8] Mark Schnittker. (2020). *How a Lock in amplifier works*. YouTube. Retrieved December 4, 2023, from <https://www.youtube.com/watch?v=R0KN3ktpvUs>
- [9] de Mendizábal, I. (2020, June 4). *Basic fundamentals of lock-in amplifiers*. All About Circuits. <https://www.allaboutcircuits.com/technical-articles/basic-fundamentals-of-lock-in-amplifiers/>
- [10] McMahill, D. (2009, February 11). *Coplanar waveguide analysis/synthesis calculator*. Coplanar Waveguide Analysis/Synthesis. <https://wcalc.sourceforge.net/cgi-bin/coplanar.cgi>

Appendix

A

```
"""
Created on Saturday Feb 27 2021

@author: Lucas Wright
for help pyvisa documentaion:
https://pyvisa.readthedocs.io/en/latest/introduction/example.htm
l
"""

#-----Library Imports-----
-----#

import pyvisa
from datetime import date
import time
import numpy as np
import random
import matplotlib.pyplot as plt
import pandas as pd
import winsound

#make the GPIB global
rm = pyvisa.ResourceManager()
instList = rm.list_resources()
#connecting to the instruments:
photonCounter = rm.open_resource('GPIB0::23::INSTR')
micEmit = rm.open_resource('GPIB0::27::INSTR')

#-----Setting Up GPIB Connectivity-----
-----#
"""fxn to return the GBIP objects (DONE)"""
def findGPIBs ():
    #get in contact with the PhotonCounter and MicEmit
    rm = pyvisa.ResourceManager()
    instList = rm.list_resources()
    print(instList)
    #connecting to the instruments:
    photonCounter = rm.open_resource('GPIB0::23::INSTR')
    micEmit = rm.open_resource('GPIB0::27::INSTR')
    return (photonCounter, micEmit)
```

```

"""This fxn will call measure() many times to build the data
list (DONE)"""
def collectData(freq_width, centered_freq, num_of_readings,
legnth_of_collection, amplitude):

    #set the amp of the MicEmit
    AmpStr = "AMPR"+str(amplitude)
    micEmit.write(AmpStr)
    centered_freq = centered_freq-200
    #generate the list of data point we want to get
pointlist[[freqLocation, length of collection],[...]:
    stepSize = freq_width/num_of_readings
    starting_freq = (centered_freq - freq_width/2)
    pointList = []

    for j in range(num_of_readings):
        pointList.append([starting_freq + (stepSize * j),
legnth_of_collection])

    #shuffle this list to decrease systematic error
    random.shuffle(pointList)

    #create an empty list to store the data:
    dataList = []
    #run a loop to gather the points:
    for freq in pointList:
        dataList.append(measure(freq[0], freq[1]))

    #here we convert to an array for funcationality

    dataListArray = np.array(dataList)

    # here we send the list on the form dataList = [[freq,
count]...]
    return (dataListArray)

""" this is where the actual measurement of the data will
happen.
Still need to figure out how to capture the photon counts for a
given time"""
def measure(freq, legnth_of_collection):

```



```

#set the microwave to correct freq\
micEmit.query("FREQ?")
freqStr = "FREQ"+str(freq)
micEmit.write(freqStr)

#sleep for a set to let micEmit start making the signal
time.sleep(.2)

# count start
photonCounter.write("cs")
#important, needs time to start the count must be larger
than 1 sec!!!
time.sleep(1)

#ask SR400 to give the photon count
count = int(photonCounter.query("QA"))
print(freq, count)
photonCounter.write("cr")
return([freq, count])

""" takes data as an array of points [[freq,count]...] plots
changes in count
normalized by the lowest value"""
def plotValues(dataList):

    #(BASIC) here we are plotting the two coloums against
eachother
    plt.scatter(dataList[:,0], dataList[:,1])
    plt.show()

def saveFile(data, name = (str(time.clock) +
str(date.fromisoformat('2019-12-04')))):
    my_df = {'freq': data[:,0],
            'photon count': data[:,1]}

    pd.DataFrame(my_df).to_csv("C:/Users/HorowitzLab/Desktop/woo
fall 24/bigersweep 2.csv", index = False)

def main():
    #User input Vars:
    #Hz
    freq_width = 0.03e9

```

```

        #Hz range 950kHz-6.075GHz
        centered_freq = 2.873e9
        #this is the number of readings we will take
        num_of_readings = 1000
        #sec
        legnth_of_collection = 0.5
        #dBm range(-110-16.5)
        Amplitude = 16.5
        micEmit.write("AMPR 16.5")

        # def collectData(freq_width, centered_freq,
num_of_readings, legnth_of_collection):
            dataList = collectData(freq_width, centered_freq,
num_of_readings, legnth_of_collection, Amplitude)
            #take data and your file name, will save to desktop folder
Lucas micWave data dataList(ARRAY)
            saveFile(dataList, "test")
            duration = 1000 # milliseconds
            freq = 440 # Hz
            winsound.Beep(freq, duration)

            plotValues(dataList)

if __name__ == '__main__':
    main()

```

B

"""

Nov 14 2023

Author: Yongwoo Park, adapted from Lucas Wright's code

Use this code to measure ODMR with lock in chopping implemented

This is a temporary script file.

"""

```

import pyvisa
from datetime import date
import time
import numpy as np
import random
import matplotlib.pyplot as plt

```

```

import pandas as pd
import winsound

#make the GPIB global
rm = pyvisa.ResourceManager()
instList = rm.list_resources()
#connecting to the instruments:
photonCounter = rm.open_resource('GPIB0::23::INSTR')
micEmit = rm.open_resource('GPIB0::27::INSTR')
photonCounter.write("DT 1E-1") # set dwell time in seconds

""" this is where the actual measurement of the data will
happen.
Still need to figure out how to capture the photon counts for a
given time"""
def measure(freq): #when pulse on

    # count start
    photonCounter.write("cs")
    #important, needs time to start the count must be larger
    #than 1 sec!!!
    time.sleep(1)

    #ask SR400 to give the photon count
    countA = int(photonCounter.query("QA"))
    countB = int(photonCounter.query("QB"))
    print(freq, countA, countB)
    photonCounter.write("cr")
    return(countA, countB)

def collectData(freq_width, centered_freq, num_of_readings,
averaging):

    #generate the list of data point we want to get
    pointlist[[freqLocation, length of collection],[]....]:
    stepSize = freq_width/num_of_readings
    starting_freq = (centered_freq - freq_width/2)

    pointList = []
    for j in range(num_of_readings):
        pointList.append(starting_freq + (stepSize * j))

    #random.shuffle(pointList)

    #create an empty list to store the data:

```

```

dataListA = []
dataListB = []

point=0
#run a loop to gather the points:
for freq in pointList:
    point = point + 1
    print(str(point)+"/"+str(num_of_readings))
    freqStr = "FREQ"+str(freq)
    micEmit.write(freqStr)

    #sleep for a set to let micEmit start making the signal
    time.sleep(.2)

    pulse_on=[]
    pulse_off=[]
    for measurement in range(averaging): # to speed up:
change to 1
        countA , countB = measure(freq)
        pulse_on.append(countA)
        pulse_off.append(countB)
        dataListA.append(pulse_on)
        dataListB.append(pulse_off)

    # here we return the frequencies and their respective A and
    B measurements
    return (pointList, dataListA, dataListB)

def processData(pointList, dataListA, dataListB):
    dataList = []
    avgListA = []
    avgListB = []

    # this loop averages the measurements per frequency and
    finds the difference for relative luminosity
    for measurement in range(len(pointList)):
        avgA = np.average(dataListA[measurement])
        avgListA.append(avgA)
        avgB = np.average(dataListB[measurement])
        avgListB.append(avgB)
        countDiff = avgA - avgB
        dataList.append(countDiff)

    return pointList, dataList, avgListA, avgListB

""" takes data as an array of points [[freq,count]...] plots
changes in count

```

```

normalized by the lowest value"""
def plotValues(pointList, dataList):

    #(BASIC) here we are plotting the two coloums against
    eachother
    plt.scatter(pointList, dataList)
    plt.show()

def saveFile(pointList, dataList, avgListA, avgListB, name =
(str(time.clock) + str(date.fromisoformat('2019-12-04')))):
    my_df = {'freq': pointList,
            'photon count': dataList,
            'A count avg': avgListA,
            'B count avg': avgListB}

    pd.DataFrame(my_df).to_csv("C:/Users/HorowitzLab/Desktop/woo
fall 24/run 4 12-10.csv", index = False)

def main():
    #User input Vars:
    #Hz
    freq_width = 0.08e9
    #Hz range 950kHz-6.075GHz
    centered_freq = 2.873e9
    #this is the number of readings we will take
    num_of_readings = 600
    #sec
    length_of_collection = 0.5
    #averaging per point
    averaging=2
    #dBm range(-110-16.5)
    Amplitude = 7
    AmpStr = "AMPR"+str(Amplitude)
    micEmit.write(AmpStr)
    #pulse stuff
    micEmit.write("MODL 1") #modulation on
    micEmit.write("TYPE 4") #pulse modulation mode
    micEmit.write("PFNC 3") #square modulation parameter
    micEmit.write("PPER 1e-5") #setting pulse period 0.01ms
    micEmit.write("PDTY 50") #pulse duty cycle 50%

    #collect data
    pointList, dataListA, dataListB = collectData(freq_width,
centered_freq, num_of_readings, averaging)

```

```
    pointList, dataList, avgListA, avgListB =  
processData(pointList, dataListA, dataListB)  
  
    #take data and your file name, will save to desktop folder  
Lucas micWave data dataList(ARRAY)  
    saveFile(pointList, dataList, avgListA, avgListB, "test")  
    duration = 1000 # milliseconds  
    freq = 440 # Hz  
    winsound.Beep(freq, duration)  
  
    plotValues(pointList, dataList)  
  
if __name__=='__main__':  
    main()
```